Managers, Wage Premia, and Worker Effort: Evidence from a Large Retailer

Job Market Paper

Latest version: `https://aneyranazarrett.github.io/research.html`

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Abstract

I study managerial monitoring and wage premia as levers to incentivize worker effort in the firm. Using administrative records from a large retailer, I show how the two jointly deter inventory shrinkage, merchandise missing from the store. Exploiting the rotation of managers across stores, I estimate each manager's effect on shrinkage and document what high-type managers do when they arrive at a store. They are not better bookkeepers. They work more shifts, including nights and weekends, spend more time on the floor with their workers, and rework the teams they inherit, yet on demographic characteristics they are indistinguishable from other managers. Leveraging a company policy that sets wages centrally by position, I document that shrinkage also falls with the wage premium. The two levers reinforce each other: the arrival of a good manager reduces shrinkage by 8 percent more where the premium is high. Good managers suspend more workers during high-wage-premium periods.

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1 Introduction

Firms across industries face a common challenge: keeping their workers' effort high over time. They have many levers to meet the challenge. Economic theory points to two main ones: monitoring and wage premia (Shapiro and Stiglitz, 1984). In the traditional model of shirking, workers put in effort because they face a probability of losing their job if they slack. A dismissed worker may be unemployed for one or more periods, and she loses a good job that is not coming back. Monitoring raises the probability that a worker who slacks is caught and dismissed. A wage premium raises the value of the current job compared to the worker's outside options, and therefore what she loses if she is dismissed. Both levers are well understood in theory. In practice, however, a firm that wants to use them faces two questions: what is the return of each lever on its profits, and how should it use them, as substitutes or as complements?

The question of whether the two levers are complements or substitutes is particularly relevant when labor markets are highly competitive, when firms compete for the same scarce workers, or when they face shocks to outside wages. In these environments, firms often end up paying low comparative premia for reasons outside their control. A firm that sets pay centrally, for example, ends up paying a low premium wherever local wages are high, or increase (Propper and Van Reenen, 2010). Another example is highly specialized labor markets, where several firms compete for the same workers and outside wages increase a lot and often. If monitoring substitutes for the premium, a good monitor can be sent to these environments to sustain effort. This is the idea behind early tests of whether firms pay higher wages where supervision is weaker (Leonard, 1987; Krueger, 1991), and firms that rely less on incentive pay for their executives rely more on board monitoring (Zajac and Westphal, 1994). If the two levers are complements, in contrast, a good monitor accomplishes little where the premium is low. Work practices raise productivity mainly when firms adopt them together (Ichniowski et al., 1997), as the theory of complementarities predicts (Milgrom and Roberts, 1995).

Calculating the return of each of the levers separately is challenging for companies. The first challenge is observing and measuring effort. Slacking at work is, by its nature, an action workers try to hide from the firm. Often firms only record the consequences of slacking, such as defects per vehicle in car manufacturing companies (Levitt et al., 2013) or food loss in restaurant chains (Detert et al., 2007). The second challenge is that when a firm's unit performs better, it is hard to separate the effect of the premia and the monitoring. The units where a firm pays higher premia are often the units where its best managers are, and where the workers put in more effort.

The literature studying these issues has focused on comparing plants or hospitals that pay different premia (Cappelli and Chauvin, 1991; Propper and Van Reenen, 2010), on whether firms pay higher wages where supervision is weaker (Leonard, 1987; Krueger, 1991), and on monitoring measured as the ratio of supervisors to workers (Coviello et al., 2022). However, understanding the dynamics of each of the levers, and the mechanics of sustaining effort, has rarely been possible, because it requires going inside the firm.

This paper examines the return of wage premia and monitoring, and whether they are complements or substitutes. I study a large retailer, a setting in which premia and monitoring can be studied separately, an effort proxy is observed, and worker behavior and composition can be analyzed in detail. I examine worker effort using inventory shrinkage, merchandise missing from the store. Shrinkage is a direct loss to the firm's bottom line. In United States retail it reached $112 billion in 2022, about 1.6 percent of sales, on the order of grocers' net profit margin (National Retail Federation, 2023; Damodaran, 2026); the most recent global survey, from 2018, puts it at a similar 1.8 percent of sales across 14 large economies (Tyco Retail Solutions and PlanetRetail RNG, 2018). Inventory shrinkage tracks worker effort; where workers put in a lot of effort at work, little goes missing from the store; where they do not, merchandise goes missing through neglect and, at the extreme, by the workers' own hand.

The retailer has more than 300 stores across Mexico and roughly 25,000 workers in a typical month, close to 100,000 over the sample period. Using the administrative data from HR, inventory and sales, I build three panels. First, I build an employee-month panel, which captures demographic data, employment history in the company, clock records, shifts, payroll, moves and their reason as well as absenteeism. It also has data on wage premium constructed using Mexico's national labor-force survey (ENOE) matching a worker in the firm to an outside worker with comparable characteristics. Second, I build a store-month panel using inventory, scanner-level data and store location characteristics. Third, a manager-month panel records which general manager runs each store, her demographics, pay, and tenure, and, from biometric time clocks, the shifts she works and how much of her time overlaps with her workers' shifts. Job descriptions and daily activity logs record what each position is expected to do, and a personnel log records every hire, promotion, transfer, and separation with their reason.

First, I study how managerial monitoring affects inventory shrinkage. In this firm there are three worker layers: a managerial layer that is in charge of monitoring, shifts, team composition, and allocation; a supervision layer; and frontline workers. I use manager rotations across stores, which are unrelated to shrinkage, to estimate an AKM model of shrinkage. Within each connected set, I define high-type (HT) managers as those with fixed effects

below the median and low-type (LT) managers as those above it. I then ask what happens to store, manager, and employee measures when an HT manager arrives at a store and what happens after she departs using a staggered event study design.

Upon the arrival of an HT manager, inventory shrinkage falls by around 7 percent. To understand how HT managers achieve this, I take the arrival and departure design and examine what they do differently upon arrival, focusing on their presence in the store, team composition, supervision, and security staffing. The arrival and departure design shows that HT managers are not better bookkeepers. They work about 1.5 more shifts per month, roughly 6 percent more than the previous manager, including nights and weekends: weekend shifts rise by about 6 percent and night shifts by about 11 percent, so the extra presence occurs when the floor is least watched. They spend more time on the floor with their teams: the share of frontline hours worked alongside a manager rises by about 3 percentage points, from a base of about 57 percent. They also rework the teams they inherit, moving workers up a tier and across areas: the monthly rate of upward moves rises by about a fifth, and moves across areas of the store rise by about 15 percent, while team size does not change. Turnover increases by about 17 percent, through quits rather than dismissals, and the workers who quit are replaced one for one. HT managers do not add security or supervisory staff, and they are indistinguishable from other managers in age, gender, education, and tenure.

Second, I study the wage premium. The company sets wages centrally by position, so a worker's premium over her local outside wage changes when outside wages in her municipality move, for reasons unrelated to her store. Comparing stores in the same municipality and quarter, I find that shrinkage falls as the premium rises: a 10 percent lower premium is associated with about 1.2 percent more shrinkage. I complement this design with a statutory change that changes the wage premia more in about a third of the company's stores. I exploit the two minimum wage areas in Mexico: border and interior areas, and statutory changes to the minimum wage in the period of study to compare stores in and out of the border. Using an event study design around the minimum-wage increases, I find that, compared to interior stores, shrinkage in border stores does not change in the first few months after an increase, but is about 4.5 percent higher in the months that follow.

Finally, I ask how managerial monitoring and wage premia interact and I look into manager behavior to see what happens when she faces a high versus a low wage premium environment. I first take the arrival design from the manager analysis and compare the arrival of a high-type manager during high- and low-wage-premium periods, defined as periods when the premium in the store's municipality is above or below its median. I find that the two levers reinforce each other: the arrival of a good manager reduces shrinkage by 8 percent more where the premium is high than where it is low. I then ask whether good managers

adjust their behavior to the wage premium, or whether their actions are the same and simply become more effective where the premium is high. Among their actions, the one that differs most clearly between the two wage environments is enforcement. High-type managers suspend more workers, imposing unpaid days off, during high-wage-premium periods, and not during low-wage-premium periods, they also seem to spend more time on the floor during high-wage premia periods.

I model worker effort by adapting the model of Shapiro and Stiglitz (1984) to the setting. The model separates two kinds of effort: hidden effort, which the firm observes only when someone is watching, such as the attention that keeps merchandise from going missing, and effort the firm can verify directly, such as attendance. A worker chooses how much hidden effort to supply, and effort is costly. The firm pays a fixed wage by position, so its only formal incentive is dismissal: a worker caught neglecting her job loses the wage premium she earns over her outside option. Whether she is caught depends on managerial monitoring, which in this setting is the general manager's presence on the floor. The deterrent is therefore a product: the probability that low effort is detected times the premium lost upon dismissal.

First, the paper contributes to a literature in operations, accounting, and organizational behavior that studies product loss in firms and links it to personnel issues and store organization. Food losses in restaurants follow how managers oversee their units (Detert et al., 2007); when a retailer changed store managers' pay to reward sales over loss prevention, managers paid less attention to shrinkage, and both shrinkage and sales rose (DeHoratius and Raman, 2007); and convenience stores that pay more than competing stores record less shrinkage and fewer cash shortages (Chen and Sandino, 2012). This paper contributes by separating the effect of the manager from the effect of the store and its workers, and by analyzing the behavior of the unit. It documents what the managers who lower shrinkage do on the floor, and shows that the same manager lowers shrinkage more where workers earn a larger wage premium.

Second, the literature studying worker effort and productivity linked to pay premia in the firm. The classic reference is the inter-plant test of Cappelli and Chauvin (1991). The paper documents a negative relationship between premia and shirking, and gives an early formal empirical test of the efficiency wage hypothesis in an inter-plant design. Propper and Van Reenen (2010) bring a similar design to the public sector. They exploit wages paid centrally in the UK. They show that pay centralization can be costly because hospitals in high-wage areas struggle to recruit and retain high-quality workers. Nurses' pay in English public hospitals is set nationally, so where outside wages are high the premium is low, and hospital quality falls, with a 10 percent higher outside wage associated with 7 percent higher death rates from emergency heart attacks. Where the premium is low, hospitals rely more

on temporary agency staff. Recent work by Emanuel and Harrington (2026) shows that a voluntary firm-wide minimum wage halves worker departures, reduces absenteeism, and raises warehouse productivity. These productivity gains fully cover the higher labor costs, so paying more can pay for itself in a private firm. My contribution is to understand how managerial monitoring and ultimately the firm's teams and organization interact in response to exogenous changes in the outside wage, outside the control of the firm. I show the dynamics of managerial response to wage premia.

The closest evidence is Coviello et al. (2022). They study salespeople paid a base rate plus commission in retail, whose pay the firm must top up to the minimum wage. A minimum-wage increase raises the productivity of the workers for whom the minimum wage is binding, but it lowers productivity where monitoring is weak. I address the same question from the opposite direction. In my setting, variation comes from managerial rotations. The personnel records measure the manager's presence and actions directly and allow the paper to measure the intensive margin as well as to take a closer look into the dynamics of the extensive margin.

Finally, the paper contributes to the growing literature that studies the roles of middle managers in firm productivity and performance (Lazear et al., 2015; Minni, 2026; Adhvaryu et al., 2026; Mollick, 2012). Related work finds that differences in management practices account for about 30 percent of productivity differences across countries (Bloom et al., 2017), that a supervisor's attention and control matter most for the productivity of the lines she runs (Adhvaryu et al., 2023), that attentive managers move workers to better-matched tasks when pollution lowers their productivity (Adhvaryu et al., 2022), that matching better managers to better lines would raise productivity (Adhvaryu et al., 2020), that middle managers who engage with their staff raise participation in the training the firm designs (Diaz et al., 2025), and that part of an individual's performance is specific to the organization where it takes place (Huckman and Pisano, 2006). My work finds that managers enforce more where the premium is high and that they provide monitoring in the store.

2 Data and Institutional Background

2.1 The Firm and the Data

I partner with a large Mexican retailer with a nationwide presence. The firm operates more than 300 stores across the country and employs around 25,000 workers in any given month. The panels cover roughly 100,000 distinct workers over the full period from January 2022 through June 2024. Central headquarters standardizes store-wide policies: physical layout, human resources, and personnel policies, including the wage by position, and the operating routines of each store.

Using the firm's administrative records and the ENOE, I build three linked panels. First, I build an employee-month panel that captures personnel records, including workers' demographic characteristics, payroll, absenteeism, worker moves, shifts, clock records, tasks, activity descriptions, and workers' wage premia. The biometric time clock is the same instrument for everyone in the store: it stamps the start and the end of every worked shift to the minute, for frontline workers and general managers alike, so the records place each person in the store day by day, nights and weekends included, and any employees' schedules can be laid over one another. The personnel-movement record behind these variables is a log of events: it registers every hire, separation with its recorded cause (resignations, dismissals, and severances enter separately), within-store reassignment across positions, tiers, and occupational areas, and every promotion and demotion. The same system carries the disciplinary record. Second, I build a manager-month database, from the same personnel system, that follows each general manager across stores over time and includes data on manager moves and the reason for each move, demographic characteristics, store presence, and absenteeism. Third, I build a store-month database that includes scanner-level sales aggregated to the month, the weekly inventory audit's shrinkage in value and physical units, store characteristics such as format and location, and store organizational structure.

2.2 Store Scale and Structure

Each store has three tiers: a managerial layer of one or more general managers (up to three in the largest formats), a middle tier of area sub-managers and supervisors, and the general workforce. Table 1 reports the distribution of sales and headcount at the store level. Each store has on average 67 workers per shift.

Table 1: Store sales and activity (store–month)

Variable	Mean	Median	Min	Max
Sales (USD)	682,726	550,828	136,785	2,087,354
Sales (units)	426,469	347,440	95,543	1,197,007
Clients (count)	47,123	42,156	3,608	118,945
Workers per store	67.3	54	1	262

Store–month observations (more than 300 stores, January 2022–June 2024). Monetary values are in USD, converted at MXN 18.5/USD, the approximate average interbank rate over the period. Inventory shrinkage is reported separately in Table 3.

2.3 Store Managers

The general manager hires and dismisses workers, assigns and reassigns them across occupational areas, promotes and moves them within the store, sets the schedule, and oversees workers. Because pay is fixed centrally by position, none of these levers runs through compensation, so her influence lies in selection, allocation, scheduling, and supervision.

2.4 Wage Premia and Effort

There are two key definitions used throughout the paper: the wage premium, which measures the rent the firm pays over the outside option, and inventory shrinkage.

Wage premia Following the efficiency-wage logic of Cappelli and Chauvin (1991), I define the wage premium as the difference between a worker's wage inside the firm and the wage a similar worker based on demographic characteristics could gain in the outside market in a given period.

$$W_{it}^{p} \;=\; w_{it} - w_{it}^{m}, \tag{1}$$

where w_{it} is worker i's firm wage in month t and w_{it}^{m} is her outside wage. I measure the outside wage by matching each worker to a reference profile in the ENOE, the national labor-force survey administered by INEGI, using gender, education, age, municipality, and quarter. Around 86 percent of worker-months find a comparable worker for all categories. Workers that do not have a comparable worker group based on all demographic characteristics receive a match relaxing one of the characteristics at a time. First I relax age, then gender, then education. After this full matching process, I have a 99 percent match rate.[1] Table 2 reports

[1]Relaxing gender, then age, then municipality accounts for a further 0.7, 0.5, and 1.2 percent of matches (so about one percent are matched at the state rather than the municipality level); education accounts for the remaining 12 percent.

the summary statistics for the full sample for wage, outside wage, and the premium as a ratio (firm wage / outside wage).

Table 2: Wages and the wage premium (worker–month)

	Mean	Median
Firm wage (USD)	476	428
Outside wage, ENOE (USD)	382	364
Wage premium (USD)	93	69
Wage premium (multiple)	1.33	1.20

Worker–month observations. The firm wage is monthly contractual pay; the outside wage is the ENOE cell wage matched on gender, education, age, municipality, and quarter (the cascade for empty cells never drops the quarter or the state). The wage premium in levels is the difference between the two; the multiple is the ratio of the firm wage to the outside wage. USD at MXN 18.5/USD.

Inventory Shrinkage The paper's central measure to capture effort is inventory shrinkage. Inventory shrinkage is defined as merchandise missing from the store that should be there according to the accounting records. Merchandise goes missing when it leaves the shelves without passing through the register, whether whole or in part. [2]

The firm audits its inventory every week, in both value and physical units. The audit compares the stock the firm's system says a store should hold with a physical count of what is on the shelf, and records the difference. The firm counts its stock daily, weekly, and monthly, so losses are caught close to when they occur. At the end of each month, the firm checks that what came into the store equals what was sold plus what went missing, so recorded losses match the measured movement of stock. Recording the inventory is a team effort, and no single person, such as the manager, records it; this is true for several retailers but especially for supermarkets. In this firm, shrinkage averages around 2.53 percent of sales a month.

This paper interprets inventory shrinkage as a key measure of effort for this supermarket. In this firm, all workers are responsible for taking care of the inventory. There are security staff and supervisors directly in charge of the inventory, but watching the stock is part of at least 90 percent of the job descriptions and activities. Shrinkage responds to two behaviors of the workforce: the effort workers devote to preventing losses on the floor (watching merchandise, handling it with care, flagging problems as they appear), and the losses workers themselves cause. This is not the first paper to show that stock loss and stock loss preven-

[2]Merchandise that is spoiled or damaged but still identifiable is logged in a separate register and is not counted as shrinkage, so the measure captures stock that is simply missing.

tion are linked to personnel policies. Food losses in restaurants follow how managers oversee their units (Detert et al., 2007); when a retailer changed store managers' pay to weight sales more heavily relative to shrinkage, managers paid less attention to preventing shrinkage, and both shrinkage and sales rose (DeHoratius and Raman, 2007); and convenience stores that pay more than competing stores record less shrinkage and fewer cash shortages (Chen and Sandino, 2012).

As additional effort outcomes, I observe turnover, the monthly rate at which workers leave the store (the records carry the reason of each separation); absenteeism, in days and as an indicator for any absence, separating non-justified from justified; and, from the scheduling system, monthly shifts and daily worked hours.

As in other large retailers, tenure is short and absence and turnover are high: median tenure is 26 months.

Table 3 reports summary statistics for shrinkage and worker behavior measures: absenteeism, turnover, monthly shifts and daily worked hours at the store and worker levels.

Table 3: Shirking measures at the store and worker levels

Variable	Mean	Median	Min	Max
Store level (store–month)				
Inventory shrinkage				
Shrinkage-to-sales (%)	2.53	2.45	0.10	10.52
Shrinkage (USD)	16,931	12,450	807	67,329
Shrinkage (units)	6,440	5,012	47	21,497
Shrinkage per worker (USD)	242	230	16	1,219
Absence (store total, days per month)				
Absence days (total)	126.3	101	0	626
Non-justified absence days	58.2	46	0	313
Turnover (% of workforce per month)				
Turnover rate (total)	6.20	5.48	0	31.25
Worker level (worker–month)				
Absence (days per month)				
Absence days (total)	1.88	0	0	31
Non-justified absence days	0.86	0	0	28
Absence (indicator)				
Any absence (= 1)	0.462	0	0	1
Any non-justified (= 1)	0.361	0	0	1
Shifts and Hours				
Monthly shifts	21.33	25	1	56
Daily worked hours	8.26	7.92	0	24

Store statistics are computed at the store–month level (more than 300 stores); worker statistics at the worker–month level, January 2022 through June 2024. USD at MXN 18.5/USD. At the store level, absence days are totaled across the store's workers, and turnover rates are separations per hundred workers. At the worker level, absence days are per-worker monthly counts and the absence indicators equal one if the worker had at least one absence of that type. *Non-justified* absence is absence without an approved justification; vacation days are reported separately and excluded here. Effort margins (shifts, hours) are recorded by the scheduling system.

Shrinkage is correlated with workforce behavior. I test pairwise correlations of shrinkage with worker behavior and with non-personnel measures such as customer traffic, commercial theft, homicide, and insecurity. I regress each store's average shrinkage-to-sales ratio on one store characteristic at a time, with both variables standardized. The workforce measures show the largest correlations: stores with lower average tenure, higher turnover and quit rates, more non-justified absence, and a larger share of new workers have higher shrinkage, with correlations between 0.17 and 0.23 in absolute value. These associations survive conditioning on store format and municipality. The correlation with customer traffic is about 0.10, and the correlations with municipal rates of commercial robbery and homicide and

with perceived insecurity are all below 0.03 in absolute value. These are consistent with shrinkage reflecting the effort of the workforce rather than the conditions of the area around the store. The figure in Appendix L shows the full set of correlations.

3 Theoretical Framework

This section formalizes how wage premia and managerial monitoring jointly deter shirking when monitoring is imperfect. I model the effort choice of a worker whose effort is care—attention a supervisor observes only imperfectly—and whose only incentive is the threat of losing a job that pays a premium over the local market wage. The model adapts Shapiro and Stiglitz (1984) and is the pure efficiency wage case of Coviello et al. (2022): fixed salaries, no pay-for-performance, an outside option equal to the market wage rather than unemployment, and dismissal as the firm's only formal incentive. The expected penalty for shirking is a *product*: the probability that a lapse is detected, which the manager sets, times the rent the worker forfeits if sanctioned, which the premium sets. The results follow the order of the paper—the monitor, the premium, their product, and who enforces it—and each proposition names the section that tests it.

3.1 Environment

Worker i of type x_i is employed at store j under manager m. Each period she chooses a *care level* $e_i \in [0,1]$—the hidden effort that keeps merchandise from going missing, including keeping time on the floor once she is there—and an *absence level* $a_i \in [0,1]$, the share of assigned shifts she shows up for. Care has consequences only if someone detects the losses it produces; absence is recorded by the time-clock and detected mechanically, whoever runs the store. Store shrinkage is $S_j = f(1-\bar{e}_j)$, with $\bar{e}_j$ average care and f strictly increasing. Costs are $c(x_i, e)$, with $c_e > 0$ and $c_{ee} > 0$, and $\kappa(a)$, increasing and convex.

The worker receives a fixed wage

$$w_i \;=\; w^m + W_i^p, \tag{2}$$

where w^m is the local market wage she can obtain immediately if she leaves[3] and $W_i^p \geq 0$ is the *wage premium* (Section 2, Equation 1). The wage does not depend on realized shrinkage. The model is stationary: the propositions are comparative statics in the standing premium, and each worker-month observation maps to the model evaluated at its current premium, treated as permanent.

The job ends through two hazards, one per margin. On the care margin, manager m monitors with intensity $\mu_m \geq 0$; a lapse by a worker exerting care e is detected with probability $q(\mu_m)h(e)$ and, in the baseline, sanctioned automatically, so she keeps her job

[3]Immediate reemployment is a normalization. With a search spell in informal work at $w_u < w^m$ and reemployment at rate λ, everything below holds with the premium augmented by $\delta \equiv (w^m - w_u)\, r/(\lambda + r)$, about three percent of the median premium under ENOE spells of one quarter.

with probability

$$\pi(e;\mu_m) \;=\; 1 - b - q(\mu_m)\,h(e), \qquad b \in (0,1),\; q > 0,\; q' > 0,\; h \geq 0,\; h' < 0,\; h'' \geq 0, \quad (3)$$

with b an exogenous separation rate, q the detection rate the manager generates, and h the worker's exposure to detection. Hence

$$\pi_e \;=\; q(\mu_m)\big(-h'(e)\big) \;>\; 0, \qquad \pi_{e\mu} \;=\; q'(\mu_m)\big(-h'(e)\big) \;>\; 0: \quad (4)$$

more care lowers the dismissal risk, and closer monitoring rewards care more strongly. The parametric form is not innocuous (Section 3.5). On the absence margin, retention is $\pi^A(a)$ with $\pi^{A\prime} > 0$ and no μ_m. Treating the two hazards as mutually exclusive within a period,

$$\Pi(e,a;\mu_m) \;=\; \pi(e;\mu_m) + \pi^A(a) - 1, \quad (5)$$

which must lie in $[0,1]$. The additive form is substantive: a multiplicative one would place μ_m inside the absence condition.

3.2 The Worker's Problem

Time is discrete. Let $V^E(x_i)$ be the value of employment and V^A that of the outside option, with flow value $u^A = \frac{r}{1+r}V^A = w^m$ since the worker can always find work at w^m. Writing the employment Bellman equation and subtracting V^A, the *employment surplus* $V(x_i) \equiv V^E(x_i) - V^A$ satisfies

$$\boxed{V(x_i) = \max_{e,\,a\,\in\,[0,1]} \left[W_i^p - c(x_i,e) - \kappa(a) + \frac{\Pi(e,a;\mu_m)}{1+r}\,V(x_i) \right].} \quad (6)$$

The market wage drops out: only the premium determines incentives. At an interior optimum the first-order conditions are

$$c_e(x_i,e^*) \;=\; \frac{\pi_e(e^*;\mu_m)}{1+r}\,V(x_i), \quad (7)$$

$$\kappa'(a^*) \;=\; \frac{\pi^{A\prime}(a^*)}{1+r}\,V(x_i), \quad (8)$$

and the surplus at the optimum is

$$V(x_i) \;=\; \frac{(1+r)\big[W_i^p - c(x_i,e^*) - \kappa(a^*)\big]}{1 + r - \Pi(e^*,a^*;\mu_m)}. \quad (9)$$

Both conditions price the surplus the worker stands to lose; only the care condition runs through the manager's monitoring.

Definition 1 (Stationary partial equilibrium of the worker)**.** *Given* (x_i, W_i^p, μ_m)*, a triple* (V, e^*, a^*) *such that (a)* $0 \leq \Pi(e, a; \mu_m) \leq 1$ *for all* (e, a)*; (b)* (e^*, a^*) *maximizes the right-hand side of* (6) *given* V*, with Kuhn–Tucker conditions* $g_e \equiv -c_e + \pi_e V/(1+r) \leq 0$ *if* $e^* = 0$*,* $= 0$ *if interior,* ≥ 0 *if* $e^* = 1$*, and likewise* $g_a \equiv -\kappa' + \pi^{A\prime} V/(1+r)$*; (c)* V *is the fixed point* (9)*; (d)* $V \geq 0$*.*

Nothing here is a firm's or a market's equilibrium: (W_i^p, μ_m) are exogenous. Write $\rho \equiv 1 + r - \Pi(e^*, a^*; \mu_m)$, $D \equiv c_{ee} - \pi_{ee} V/(1+r)$, $D_A \equiv \kappa'' - \pi^{A\prime\prime} V/(1+r)$, and let $R = W \times [\mu_{LT}, \mu_{HT}]$ be a rectangle of premiums and monitoring intensities on which (e^*, a^*) is interior, with c and h three times and κ, π^A twice continuously differentiable.

Lemma 1. *Under* (3) *and* $D_A > 0$*: (i)* V *exists and is unique, the care problem is single-peaked, and* $D > 0$*; (ii) the care condition is* $M(e^*) = q(\mu_m)\, V/(1+r)$ *with* $M(e) \equiv c_e(x_i, e)/(-h'(e))$ *and* $M' > 0$*, so care is a strictly increasing transform of the expected-penalty product; (iii)* V *is strictly increasing in* W_i^p*, and* $V > 0$ *at an interior optimum requires* $W_i^p > c(x_i, e^*) + \kappa(a^*)$*.*

Proofs are in Appendix A.

3.3 Linking the Model to the Setting

The monitor. The argument of (7) is the monitoring the worker actually faces in store j in period t, $\mu_{mjt} = \bar{\mu}_m + u_{mjt}$, with $\bar{\mu}_m$ portable. The arrival designs classify managers by their AKM shrinkage fixed effect (Section 4), high-type (HT) below the median of their connected set; Section 5 grounds the label in conduct: high-type managers work more shifts, overlap more of their workers' hours, and their workers keep better time while they are on the floor. Monitoring is not an apparatus the manager purchases but her presence on the floor, which proxies μ_{mjt} itself; nothing in her problem re-optimizes it against the local rent, so her actions should not vary with the premium while the workers' response does (Section 7.3). Where label and realized presence disagree as moderators (Section 8.1), the model sides with the realized value.

The premium. In Shapiro and Stiglitz (1984) the punishment for shirking is unemployment; here the market always offers w^m, what dismissal destroys is the premium, and (6) makes the premium the whole of the incentive. Workers earn on average about 1.33 times their outside wage (Section 2). Because the contract wage is set centrally by position and

moves rarely, movements in the local outside wage pass through one-for-one to the premium, the variation the two-way fixed-effects design of Section 6 exploits.[4] A statutory minimum-wage hike (Section 8) moves the premium in two phases with opposite signs: on impact it raises the contract wage of the workers the floor covers, the case Coviello et al. (2022) study; as the local outside wage catches up against a scale that does not move, the premium compresses, the case they set aside as demotivating. Proposition 2 signs both; the stationary model does not date the second.

The no-shirking condition. In the binary case of Shapiro and Stiglitz (1984)—$e \in \{0, \bar{e}\}$, absence suppressed, retention $1-b$ with effort and $1-b-q$ without—the surplus of a worker who exerts effort is $V_H = (1+r)(W_i^p - \bar{e})/(r+b)$, a one-period deviation saves $\bar{e}$ and exposes V_H to detection risk q, so effort is optimal if and only if

$$W_i^p \;\geq\; \bar{e} + \frac{(r+b)\,\bar{e}}{q} \;\equiv\; \widehat{wp}, \tag{10}$$

the effort cost, which a rent gross of effort must cover, plus the no-shirking mark-up. Read as the firm's menu, the condition makes the two instruments *substitutes*: the same care can be sustained with more monitoring and less premium. That is the null the arrival design of Section 7 rejects in levels.

3.4 Results

Proposition 1 (The monitor disciplines the hidden margin)**.** *Let (e^*, a^*) be interior. Under higher monitoring intensity μ_m:*

(i) Care is higher at any given premium, $\partial e^*/\partial \mu_m > 0$, and shrinkage lower.

(ii) Absence does not fall. Each margin's response to the manager is its response to the premium rescaled by a detection stake,

$$\begin{aligned} \frac{\partial a^*}{\partial \mu} &= -\frac{q'(\mu)\,h(e^*)\,V}{1+r} \cdot \frac{\partial a^*}{\partial W^p} \;\leq\; 0, \\ \frac{\partial e^*}{\partial \mu} &= \frac{q'(\mu)\,V}{1+r} \cdot \frac{r+b+1-\pi^A(a^*)}{q(\mu)} \cdot \frac{\partial e^*}{\partial W^p} \;>\; 0, \end{aligned} \tag{11}$$

so a stricter monitor moves absence only through the residual exposure $h(e^*)$ of a

[4] The treatment of monitoring parallels Coviello et al. (2022): detection depends on effort when monitored, termination follows detection, monitoring is exogenous. Their μ is the share of workers monitored in a store; here μ_m is the intensity embodied in the manager.

compliant worker, zero in the benchmark $h(e^*) = 0$, while her effect on care runs through the slope $-h'(e^*)$ and is bounded away from zero.

(iii) With an unanticipated shock $\varepsilon \sim G$ to the outside option, realized after actions, quits rise with μ_m in proportion to $g(V)\, q'(\mu) h(e^*) V/\rho$: only among workers who still carry exposure. The dismissal rate of a compliant worker, $q(\mu)h(e^*)$, stays near zero.

Part (i) uses the form (3), not only the signs (4): a stricter monitor also trims the surplus of a worker with exposure, and under (3) the deterrence channel dominates.[5] Sections 4 and 5 test the proposition: shrinkage falls when a high-type manager arrives and climbs back when she leaves; turnover rises; and the test of (ii) is the arrival design on non-justified absence (Section 7, Appendix H), not the comovement of manager effects across outcomes, which validates the label and is silent on the mechanism.

Proposition 2 (The premium disciplines both margins)**.** *At an interior optimum:*

(i) Care rises, $\partial e^*/\partial W^p = \pi_e/(\rho D) > 0$, and shrinkage falls.

(ii) Absence falls, $\partial a^*/\partial W^p = \pi^{A\prime}/(\rho D_A) > 0$, and time on the job rises.

(iii) The dismissal hazard $1 - \Pi$ falls, and with the shock of Proposition 1(iii) quits fall as well.

Sections 6 and 8 test it: shrinkage, absence, and turnover fall and shifts and hours rise with the premium, and the statutory compression raises shrinkage. Two assumptions sign what follows.

Assumption 1 (Compliant-care detection)**.** *(a) The care hazard is* (3)*. (b) At every optimum in R, the residual exposure of compliant care is small relative to its slope:*

$$h(e^*)\, \pi^{A\prime}(a^*)\, \kappa'(a^*)\, D \;<\; \big(-h'(e^*)\big)\, c_e(x_i, e^*)\, \big(r + b + 1 - \pi^A(a^*)\big)\, D_A. \tag{12}$$

Assumption 2 (Curvature regularity)**.** *$M(e) = c_e(x_i, e)/(-h'(e))$ is log-concave,* $(\log M)'' \leq 0$.

Part (b) is "$h \approx 0$ at compliant care" as an inequality; the benchmark $h(e^*) = 0$ satisfies it strictly. Assumption 2 disciplines the third-order terms the cross-partial of an optimum drags in and holds whenever c_e and $-h'$ have constant elasticities.

[5] Under $\pi_e \geq 0$, $\pi_{e\mu} \geq 0$ alone, $\partial e^*/\partial \mu = \frac{V}{(1+r)D}[\pi_{e\mu} + \pi_e \pi_\mu/\rho]$ carries the unrestricted sign of π_μ: with $c = e^2/2$, $r = 0.05$, $W^p = 0.05$, $\pi = 0.8 + \mu e - 0.4\mu^2$, at $\mu = 0.9$ the solution is $e^* \approx 0.084$, $\pi^* \approx 0.55$, and $\partial e^*/\partial \mu \approx -0.014$. The form (3) is a parametric member of the monitoring order of Coviello et al. (2022, Definition 2), which rules such cases out.

Proposition 3 (The premium and the monitor are complements)**.** *Let $e^*(W^p, \mu)$ solve* (7) *on R under Assumption 1(a). Then:*

(i) ***(One product does all the work.)*** *Optimal care satisfies*

$$M(e^*) \;=\; q(\mu)\,\frac{V}{1+r}, \qquad M' > 0, \tag{13}$$

with slopes

$$\frac{\partial e^*}{\partial W^p} = \frac{\pi_e(e^*;\mu)}{\rho\, D} \;>\; 0 \qquad \textit{and} \qquad \frac{\partial e^*}{\partial \mu} = \frac{q'(\mu)\big(-h'(e^*)\big)\big(r+b+1-\pi^A(a^*)\big)\,V}{(1+r)\,\rho\, D} \;>\; 0. \tag{14}$$

(ii) ***(Local complementarity.)*** *Under Assumptions 1–2,*

$$\frac{\partial^2 e^*}{\partial W^p\,\partial \mu} \;>\; 0 \qquad \textit{on } R. \tag{15}$$

(iii) ***(The premium amplifies the monitor.)*** *For every $W^p \in W$, $\Delta_\mu e^*(W^p) \equiv e^*(W^p, \mu_{HT}) - e^*(W^p, \mu_{LT})$ is strictly positive and strictly increasing in W^p: for $W^p_L < W^p_H$ in W,*

$$\Delta_\mu e^*(W^p_H) - \Delta_\mu e^*(W^p_L) \;=\; \int_{W^p_L}^{W^p_H}\!\!\int_{\mu_{LT}}^{\mu_{HT}} \frac{\partial^2 e^*}{\partial W^p\,\partial \mu}\, d\mu\, dW^p \;>\; 0. \tag{16}$$

Propositions 1(i), 2(i), and 3 are three readings of (13): the derivative of the product in the detection rate, in the rent, and their cross-partial. The premium's proportional grip on the surplus is $\partial \log V/\partial W^p = 1/(W^p - c - \kappa)$, and a stricter monitor raises the cost of the actions the worker chooses, so the same peso moves the surplus by more in proportion; the sign is established under Assumptions 1(b) and 2, not by the product alone. Part (iii) is what Section 7 tests cell by cell—the reinforcement Coviello et al. (2022) find from the opposite direction.

Enforcement. In the store the sanction is the manager's act—a suspension or a dismissal she must file, argue, and staff around—and it costs her.[6] What it buys her is deterrence, which runs only through what the sanction confiscates: a worker with no rent to lose is indifferent to losing the job. Suppose she sanctions a detected lapse if and only if the rent it

[6] A dismissal confiscates the rent V; a suspension without pay confiscates a share of it. The threshold logic is the same, and suspensions are the sanction the records observe (Section 7.3).

confiscates is at least $\bar{v} > 0$. Writing $s \equiv \mathbf{1}[V \geq \bar{v}]$, with V the surplus enforcement sustains, the care hazard becomes $\pi = 1 - b - s\, q(\mu)h(e)$: the detection technology is the manager's presence, but the threat behind it is exercised only where there is rent to confiscate.[7]

Proposition 4 (Enforcement follows the rent)**.** *Under the sanction rule and Assumption 1(a):*

(i) ***(Credibility threshold.)*** *There is a premium threshold $\overline{W}(\mu)$, defined by $V(\overline{W}, \mu) = \bar{v}$, such that enforcement is credible if and only if $W^p \geq \overline{W}(\mu)$. Below it $\pi_e = 0$: care sits at its lower corner and responds neither to the premium nor to the monitor.*

(ii) ***(Sanctions on impact.)*** *When a stricter monitor replaces a laxer one, $\mu_L \to \mu_H$, at the care level workers chose under the old monitor the sanction rate rises by $\left[q(\mu_H) - q(\mu_L)\right] h(e^*_L)$ where enforcement is credible under the new monitor, $W^p \geq \overline{W}(\mu_H)$, and by nothing where it is not.*

(iii) ***(Extensive margin.)*** *Above the threshold Proposition 3 holds unchanged; across it, the arrival effect on care rises from zero to $\Delta_\mu e^*(W^p) > 0$. The share of a store's workers above the threshold rises with the store's premium, so the store-level arrival effect rises with the premium through both margins.*

Proposition 4 is the mechanism Section 7.3 tests. The arrival of the same manager should show the same presence in every wage environment, more sanctions where the premium is high and none where it is low, and a larger fall in shrinkage where the premium is high. The signed object is the differential in sanctions on impact; the stationary sanction rate $q(\mu_H)h(e^*_H)$ is not signed. The same rule explains why no premium rescues a store from a weak monitor: with $q(\mu)$ near zero the sanction rate is near zero at any rent (Section 7, low-type panel).

Corollary 1 (Aggregation)**.** *Let store j employ N_j workers with premiums $W^p_{ij} = \overline{W^p}_j + \eta_i$, the joint distribution of (x_i, η_i) fixed, a common μ_j, every (W^p_{ij}, μ_j) in R, and $S_j = \alpha_j + \frac{1}{N_j}\sum_i \ell(1 - e^*_{ij}) > 0$ with ℓ linear and increasing. Under the hypotheses of Proposition 3, $\partial S_j / \partial \overline{W^p}_j < 0$, $\partial S_j / \partial \mu_j < 0$, $\partial^2 S_j / \partial \overline{W^p}_j \partial \mu_j < 0$, and the same signs hold for $\log S_j$.*

With convex losses, $f'' > 0$, the corollary requires $f''\, \bar{e}_{W^p}\, \bar{e}_\mu < f'\, \bar{e}_{W^p \mu}$; it also holds composition fixed, which the team evidence of Section 5 licenses.

[7] The rule is a commitment evaluated at the surplus under enforcement. Evaluated lapse by lapse it leaves a band of premiums, between the surplus a non-enforcing manager's workers hold and $\bar{v}$, where the only consistent policy sanctions with some probability and the threshold of part (i) becomes a ramp.

Corners. Section 8.1 reads a pattern the interior results do not predict: a marginal compression of the premium raises shrinkage only where the manager is not on the floor. On R, by (15), an interior compression removes *more* care where monitoring is high. The insured store is a corner. By Definition 1(b) and Lemma 1(ii), care is at full compliance, $e^* = 1$, if and only if

$$\begin{aligned} \Gamma(W^p, \mu) &\equiv \frac{q(\mu)\, V^{(1)}(W^p, \mu)}{1+r} - M(1) \;\geq\; 0, \\ V^{(1)}(W^p, \mu) &\equiv \frac{(1+r)\big[W^p - c(x_i, 1) - \kappa(a^*)\big]}{r + b + q(\mu)h(1) + 1 - \pi^A(a^*)}, \end{aligned} \tag{17}$$

with $V^{(1)}$ the surplus at full care and a^* solving (8) at $V = V^{(1)}$.

Proposition 5 (Saturation and marginal insurance)**.** *Under Assumption 1(a), with $D_A > 0$ and $V^{(1)} > 0$:*

(i) Γ *is strictly increasing in W^p and in μ, so there is a threshold $\bar{\mu}(W^p)$, decreasing in W^p, with $e^*(W^p, \mu) = 1$ if and only if $\mu \geq \bar{\mu}(W^p)$.*

(ii) ***(Marginal insurance.)*** *Fix W^p and $\mu_L < \bar{\mu}(W^p) < \mu_H$. For every sufficiently small compression $dW^p < 0$,*

$$\begin{aligned} e^*(W^p + dW^p, \mu_H) - e^*(W^p, \mu_H) &= 0, \\ e^*(W^p + dW^p, \mu_L) - e^*(W^p, \mu_L) &= \frac{\pi_e}{\rho D}\, dW^p + o(dW^p) \;<\; 0. \end{aligned} \tag{18}$$

(iii) If instead both (W^p, μ_L) and (W^p, μ_H) lie in R under Assumptions 1–2, the ordering reverses: the same compression removes more care under the strict monitor.

Section 8.1 tests Proposition 5. It is not the marginal form of the complementarity—part (iii) orders the stores the other way—but a statement about where each store sits relative to the corner: a compression damages the unwatched store because its care margin is interior and spares the watched one because a small loss of rent leaves the slack Γ positive. Part (i) says which stores should be saturated, those with a high premium *and* a strict monitor, and a compression large enough to push Γ below zero would end the insurance. The same logic on the absence margin, $g_a(1) \geq 0$, insures the attendance of a worker at full compliance.

3.5 Discussion

Workers are risk neutral throughout. Appendix A.2 records where the data refine the model: monitoring is an attribute embodied in the manager rather than a technology the firm

buys; the complementarity lives in the worker's first-order condition rather than in the manager's conduct, which barely responds to the premium; and shirking splits into a care margin that must be detected and an attendance margin that records itself, so that the level complementarity and the marginal insurance of the watched store are two comparative statics of one condition. Three comments.

The detection technology is load-bearing. The form (3) delivers concavity in care by construction and carries the property the two-margin results rest on: a compliant worker has little to fear from a watchful manager, so monitoring changes the return to care, not the survival odds of the compliant; without it the monitoring comparative static can reverse. Assumption 1(b) is stated at the optimum: by Equation (32) of Appendix A.1 it is equivalent to a positive induced-retention response, $\beta \equiv \pi_e \, \partial e^*/\partial \mu + \pi^{A\prime} \, \partial a^*/\partial \mu > 0$, which under linear exposure $h = h_0(1 - e)$ holds whenever compliant care is close to full or the absence hazard is flat at the optimum. Exact invariance of absence is a limit, not a specification: $h \equiv 0$ would erase the care margin too, so the model delivers the bound in (11), which vanishes at fixed slope only as $e^* \to 1$, the saturation region of Proposition 5. $D_A > 0$ is assumed, with sufficient conditions parallel to Assumptions 1–2 of Coviello et al. (2022).

What the stationary model does not predict. Timing: the model compares steady states and does not date the compression's lag (Sections 8 and 8.1); a partial-adjustment extension, $e_{t+1} = (1 - \lambda)e_t + \lambda \, e^*(W_t^p, \mu_t)$, delivers it, and its impact period is where Proposition 4(ii) lives. Team production: the hazard $q(\mu)h(e_i)$ assumes detection identifies individual negligence; were losses attributable only to the store, care would be a public good within the team. Corollary 1 holds N_j fixed.

The low-type pattern. A low-type arrival raises shrinkage in both wage environments with no significant difference between them (Section 7). The model predicts the sign, by (14), not the absence of scaling: on the interior, (15) makes the damage from a weak monitor larger where the premium is high. The flat pattern is descriptive, consistent with imprecision or with the low-premium store sitting at the lower corner under the weak monitor. What the model does license is the weaker statement: a premium disciplines the shrinkage margin only through a monitor who enforces (Proposition 4).

4 Manager Effects on Shrinkage: AKM Decomposition

I study the general manager's effect on inventory shrinkage using the firm's rotation of managers across stores. I proceed in three steps. First, I estimate a two-way fixed-effects model following Abowd et al. (1999) (AKM), leveraging manager rotations that are unrelated to store shrinkage. Second, I classify managers as high- or low-type by their estimated fixed effects. Third, I take the classification and do an arrival and departure design by manager type (HT/LT managers). In particular, I use an event study design defining the event as the arrival/departure of an HT/LT manager to a store and test what happens to shrinkage, manager presence and behavior, team composition and supervision apparatus in the store.

4.1 Estimating Manager Fixed Effects on Inventory Shrinkage

For the first step, I use an AKM model on inventory shrinkage and calculate the manager and store fixed effects leveraging manager moves. I estimate the following model:

$$Y_{jt} \;=\; \theta_{m(j,t)} \;+\; \psi_j \;+\; \delta_t \;+\; \varepsilon_{jt}, \tag{19}$$

where Y_{jt} is log shrinkage value in store j and month t, $\theta_{m(j,t)}$ is a fixed effect for the general manager running the store, ψ_j absorbs permanent store characteristics, and δ_t absorbs seasonality and firm-wide shocks. Following Card et al. (2013), the error decomposes as

$$\varepsilon_{jt} \;=\; \eta_{m(j,t),j} \;+\; \xi_{jt} \;+\; \nu_{jt}, \tag{20}$$

a manager–store match component, a store component, and transitory noise. Identification requires that manager–store assignment be mean-independent of ε_{jt}: managers may sort on the permanent components, which the fixed effects absorb, but not on the match or on transitory shocks (Appendix B states the moment conditions).

The first concern for identification is that manager assignments are related to inventory shrinkage and therefore that shrinkage is correlated with the manager-store match specific component. Conversations with the partner firm and three formal tests give evidence against the concern. First, although the firm audits shrinkage weekly, it is not a key performance indicator in manager assignment: human resources does not track it systematically, and, by the firm's own descriptions of manager reassignments, placement is not directed by the shrinkage of the destination store. In particular the firm does not send its best managers to the stores with the most shrinkage. Second, manager assignment does not seem to be directly correlated with something that might be strongly correlated with shrinkage. The movement

records themselves show coordinated rotation across stores for manager development. Junior store managers are rotated across formats for store development or to replace other managers leaving stores. Table 13 (Appendix B) classifies the roughly 520 between-store moves: one in five manager moves is a direct swap between two stores, and one in three fills a vacant store, and the moves cluster in waves: a handful of months absorb the largest share, consistent with coordinated centralized rotations rather than case-by-case reactions to store conditions. Managers do not move toward stores with higher or lower shrinkage. For each move, I compare the shrinkage of the store the manager leaves with the shrinkage of the store she arrives at, measured before she arrives. On average, the destination's pre-move shrinkage exceeds the origin's by 2.6 percent, a difference not distinguishable from zero ($p = 0.27$).

Third, the moves pass two tests of exogenous mobility. Appendix Figure 12 tests it directly, with the event-study test of Card et al. (2013). I rank stores by quartile of their average shrinkage, isolate the managers who move across quartiles, and follow each manager across her move: the outcome at every date is the shrinkage of whatever store she is running, so at the move the store changes and the manager does not. If assignment were unrelated to the error term, a mover's outcome should jump to the destination store's level on impact—the jump measures the store, not the manager—with no differential trend before the move and with upward and downward moves mirroring each other. The figure shows this pattern: movers from bottom- to top-quartile stores step up to the destination level on impact and the reverse moves fall symmetrically; within-quartile moves are flat, so the step is not an artifact of moving itself; and trajectories are flat within origin quartile before the move, so movers were not selected on a diverging shrinkage path. The stores, not the managers, carry the level differences the figure displays, which is what the decomposition assumes when it separates the two.

The second test uses the symmetry that mean-independence imposes on the match component of Equation (20). If managers moved to stores where their particular match is favorable, moving up and moving down the store-quality ladder would both look advantageous, and the gains from upward moves would exceed the losses from downward ones. Appendix Figure 13 plots the average residual change for each upward quartile pair against the corresponding downward pair; symmetry implies the -45-degree line. The points cluster around that line, and no pair deviates in the direction implied by match-based sorting: what a manager gains by moving up a quartile, another loses by moving down it. Together with Figure 12, this is what allows the fixed effect to be read as the manager's own contribution rather than as the effect of the store to which she was assigned.

I also address limited-mobility bias. A majority of general managers are movers: 55.0 percent run at least two different stores in the sample, 25.4 percent run three or more, and

91.7 percent of stores receive at least one mover (Table 12, Appendix B)—high relative to the matched employer–employee panels on which AKM methods were developed, where mover shares of 12, 25, and 35 percent are typical (Andrews et al., 2012; Card et al., 2013; Alvarez et al., 2018). Manager and store effects are identified within connected sets of stores linked by moves; the largest covers about three-quarters of the panel, and every decomposition below normalizes within its set. The remaining concern is statistical rather than economic: fixed effects estimated from finite mover counts are noisy, and the noise inflates plug-in variance components. I address it with the leave-out estimator of Kline et al. (2020) (KSS).

Table 4: AKM variance decomposition with limited-mobility-bias corrections (log shrinkage value)

	Baseline	*Andrews*	*KSS*
$\widehat{\text{Var}}(\theta)$	0.049	0.009	0.016
$\widehat{\text{Var}}(\psi)$	0.287	0.243	0.251
$\widehat{\text{Cov}}(\theta, \psi)$	−0.014	0.002	−0.002
$\widehat{\text{Corr}}(\theta, \psi)$	−0.120	0.044	−0.038

Largest connected set of the general-manager panel. θ = manager fixed effect, ψ = store fixed effect; observation-weighted variances and covariances. *Baseline*: OLS variance of the estimated fixed effects. *Andrews*: homoskedastic correction of Andrews et al. (2008). *KSS*: leave-out estimator of Kline et al. (2020).

Table 4 reports the variance decomposition with and without the correction. In the plug-in decomposition, manager fixed effects account for 49.8 percent of the variance of log shrinkage value and store fixed effects for 42.4 percent—covariance shares, $\text{Cov}(y, \cdot)/\text{Var}(y)$, which allocate the sorting covariance and so need not mirror the raw variance components the table reports. [8]

The leave-out correction cuts the manager component to roughly a third of its plug-in value—$\widehat{\text{Var}}(\theta)$ falls from 0.049 to 0.016. The corrected component remains economically large: a manager one standard deviation better is associated with about 12 percent lower shrinkage ($\sqrt{0.016}$), holding the store and the month fixed. Appendix B plots the estimated fixed effects themselves; those plots convolve true effects with sampling noise, which is why the inference rests on the corrected components of Table 4 rather than on raw dispersion.

The classification is also insensitive to the estimation error that the correction quantifies. Managers are labeled by a median split within connected set, so the label depends only on

[8] Every manager claim in the paper rests on the *KSS* column; the plug-in shares are quoted only to show what the correction changes.

ranks, and shrinking each fixed effect toward its cell mean in proportion to its noise reorders very few of them: an empirical-Bayes correction under the plug-in reliability reclassifies fewer than one percent of managers, and a correction under the reliability implied by Table 4 itself—roughly one third—reclassifies under two percent. Re-estimating every design that uses the label under the corrected classifications leaves the results unchanged: the arrival effect, the enforcement response, the presence response, and the complementarity differential retain 88 to 93 percent of their magnitudes at the same significance levels.

The decomposition assigns the manager a first-order role in shrinkage, of the same order as every permanent characteristic of the store combined. It is an accounting result, built on fixed effects; the sharpest confirmation of the economic content of the manager component is in the next section, which leverages the firm's rotation of managers directly and asks whether shrinkage moves when the manager moves, in an arrival-and-departure design.

5 What Do Good Managers Do?

A central question that the paper answers is what good managers are doing across stores. This section exploits the high-frequency data on managers' actions—the minute-level time clock, the personnel record, and the store's operational panel—to ask what the managers whose fixed effect reduces shrinkage do differently inside the store. This section performs two validations of the fixed effects and then takes the arrival design across manager behavior.

Classifying managers I classify general managers into two types based on their AKM fixed effects for log shrinkage value, estimated in Section 4. A manager is high-type (HT) if her fixed effect falls below the median of the manager-effect distribution within her connected set; otherwise, she is low-type (LT).[9]

Design I use an arrival design throughout the section. An *arrival* occurs the first time a store comes under a manager of the given type for two consecutive months, and the first of those months is the event date; the two-month rule screens out temporary covers and end-of-month payroll overlaps. The period before the arrival is the reference period and we call it the handover month. A *departure* is the mirror image: the first month the store is no longer under that manager after the first arrival. We use a within-store treatment and test for robustness with staggered treatment designs. For a store-month outcome Y_{jt} I estimate

$$Y_{jt} \;=\; \sum_{\substack{k=-8 \\ k\neq -1}}^{8} \beta_k \, \mathbf{1}[t - E_j = k] \;+\; \varphi_j \;+\; \lambda_{m(t)} \;+\; \xi_{y(t)} \;+\; \varepsilon_{jt}, \tag{21}$$

where E_j is store j's event date, φ_j is a store fixed effect, $\lambda_{m(t)}$ and $\xi_{y(t)}$ are calendar-month and year effects, and the endpoint dummies bin event times beyond ± 8 months. Standard errors are clustered at the store level. A companion difference-in-differences replaces the event-time dummies with a single $\text{Post}_{[0,8]}$ indicator for the first nine months after the event; that coefficient is the one reported in every table cell.

5.1 Validating the Classification of HT Managers

First, I validate the HT management classification; if the AKM fixed effect carries economic meaning, the arrival of an HT manager should reduce shrinkage. Using the event study design from equation 21, I find that shrinkage falls 6.6 percent over the nine months following a

[9] A lower fixed effect means less shrinkage under her management, so high-type managers are the better half of the distribution.

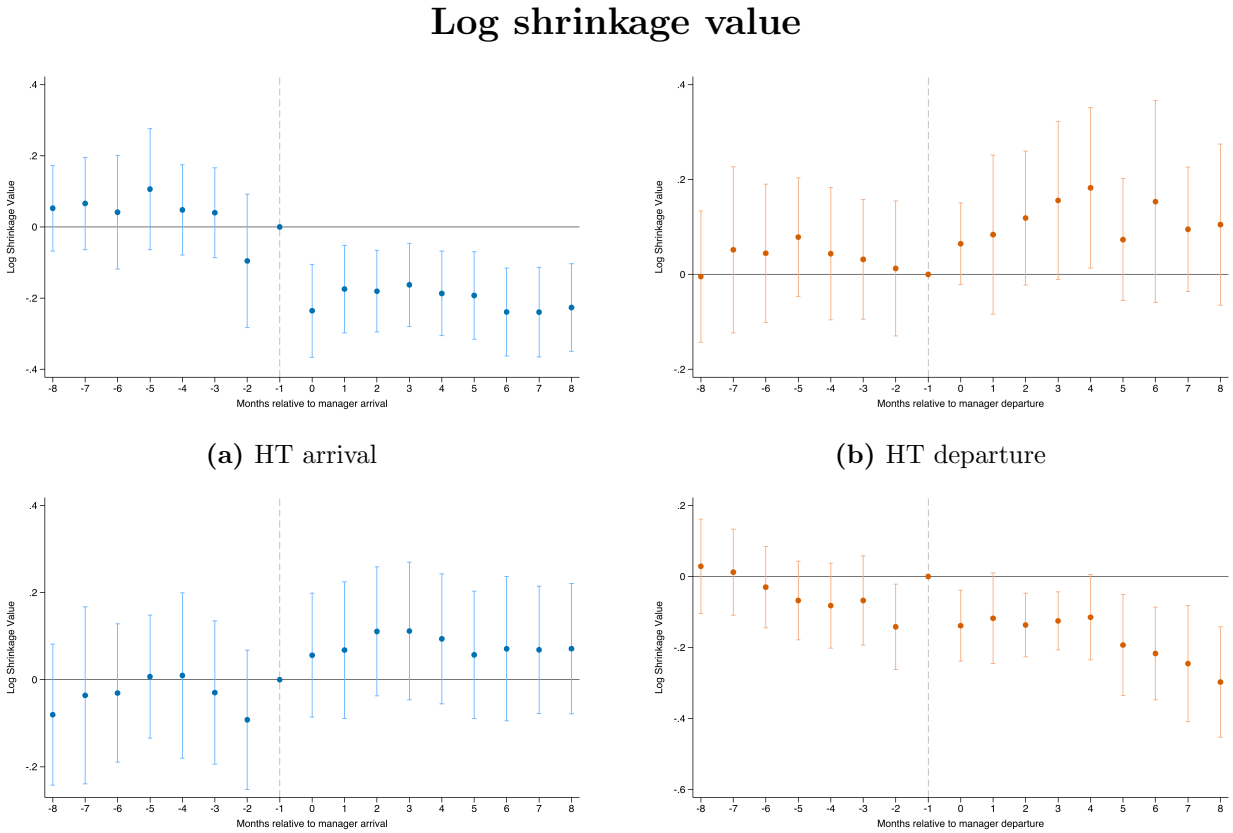
high-type manager's arrival and rises 6.7 percent after her departure, reversing essentially the whole arrival reduction. The low-type designs mirror these results: shrinkage rises by 5.0 percent when a low-type manager arrives and shows no significant change when she departs. Appendix C reports these same four designs for shrinkage-to-sales, with the same pattern.

Table 5: Shrinkage on manager arrivals and departures

	Log shrinkage value
Arrival of a high-type manager	-0.0683^{***}
	(0.0146)
Departure of a high-type manager	0.0653^{*}
	(0.0336)
Arrival of a low-type manager	0.0489^{***}
	(0.0176)
Departure of a low-type manager	-0.0343
	(0.0305)
Mean	12.43
Median	12.35

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a difference-in-differences regression of the store-month outcome on a single post indicator equal to one in the first nine months after the event (event months 0 to 8) and zero in every other month, with store, calendar-month, and year fixed effects and standard errors clustered at the store, on the log value of inventory losses. Mean and Median are of the log outcome. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

Under staggered timing, the pooled coefficient in Table 5 provides a conservative measure: stores treated earlier act as controls for stores treated later while their own shrinkage is still depressed, which attenuates the estimate (Goodman-Bacon, 2021). However, restricting the same regression to the event window yields a much stronger result: shrinkage falls by 17 percent after a high-type arrival, aligning closely with the depth of the event-study path.



(a) HT arrival

(b) HT departure

(c) LT arrival

(d) LT departure

Figure 1: Log shrinkage value around the four manager transitions

Notes: Each panel plots the coefficients β_k from an event-study regression of the store-month outcome on event-month indicators, with store, calendar-month, and year fixed effects, for one of the four transition designs, estimated on that design's own event sample: (a) the arrival of a high-type manager, (b) the departure of a high-type manager, (c) the arrival of a low-type manager, and (d) the departure of a low-type manager. A manager is classified as high-type (HT) if her fixed effect in the decomposition of log shrinkage value into manager, store, and month fixed effects, lies below the median of the connected set—her stores lose less merchandise under her than under the median manager—and low-type (LT) otherwise. The outcome is the store's log shrinkage value in the month; the unit of observation is the store-month. The month before the event, the handover month ($k = -1$), is the omitted category. Store, calendar-month, and year fixed effects; ever-treated stores of each design. The vertical line marks the handover month. Bands are 95 percent confidence intervals with standard errors clustered by store.

Validating Management with Other Shirking Margins A manager's shrinkage fixed effect could reflect careful recordkeeping rather than management. Conversely, an HT manager in shrinkage can capture a broader quality across outcomes. To test this, I re-estimate the decomposition of Equation (19) separately for five additional outcomes—two alternative shrinkage definitions, turnover, and two non-justified-absence measures, the latter three recorded by payroll and the time clock, outside any manager's inventory paperwork—and compare each manager's fixed effects across outcomes. I find that the fixed effects from shrinkage are highly correlated with the fixed effects on other organizational characteristic

measures. In other words, HT managers on shrinkage are also high-type managers in other measures.

Figure 2 plots each manager's fixed effect in every outcome against her fixed effect on every other one, and every pairwise correlation is positive: the three shrinkage-based effects are strongly correlated (0.63–0.85), and the shrinkage-value effect moves together with the manager effects in turnover (0.38) and in non-justified absence (0.65–0.82). High-type managers carry better fixed effects in every other outcome—strongly so for the shrinkage normalizations and turnover ($p < 0.001$), more weakly for the two absence measures ($p = 0.06$–0.09). If the fixed effect rewarded recordkeeping rather than conduct, it would not travel to margins the manager does not record. What the label captures is a broad dimension of managerial quality, in line with the manager-specific productivity effects of Lazear et al. (2015) and Bertrand and Schoar (2003), and this supports reading the arrival effects that follow as the effect of a better manager. The arrival effect is also graded rather than binary—it scales with how much the arriving manager improves on the incumbent one and vanishes for arrivals that are not upgrades—a dose response and novelty placebo reported in Appendix D.2.



Figure 2: Manager fixed effects across six outcomes

Notes: Each panel plots one point per manager, her fixed effect in one outcome (horizontal axis) against her fixed effect in another (vertical axis). Each fixed effect comes from a separate decomposition of that outcome into manager, store, and month fixed effects, estimated on the connected set, and is standardized to standard-deviation units. The six outcomes are log shrinkage value (first row and column), shrinkage per sales, shrinkage per worker, turnover, and two measures of non-justified absence, one from payroll and one from the time clock. Higher values mean more shrinkage, turnover, or absence. Pairwise Pearson correlations range from 0.38 (turnover) to 0.85 (across shrinkage definitions).

An additional check asks whether the classification is picking up something a personnel

file already shows—that the high-type managers are simply the senior ones, for instance, or the better paid. It is not. High- and low-type managers are statistically indistinguishable on the demographic characteristics that the firm records: gender (a difference of 2.4 percentage points, $p = 0.36$), schooling (0.16 years, $p = 0.39$), and age (0.98 years, $p = 0.07$). Tenure does not differentiate them either; if anything, high-type managers have been at the firm slightly less time, not more, and they are paid the same (Appendix D.6 reports every comparison). On paper, the good manager is ordinary; what the fixed effect captures is not a credential the firm already prices, which is precisely why identifying good managers from personnel records alone is hard.

5.2 Co-Presence and not a larger monitoring apparatus: Management as Monitoring

What does a good manager do differently once she arrives? The most direct way a manager can make lapses more likely to be caught is to watch the floor herself. She could instead put more watching into the store, by adding loss-prevention, supervisory, or security staff. I test both, and the answer is that she does the former and not the latter: after a high-type arrival the manager is present more and is on the floor with her workers more, while the store's monitoring staff does not change.

The first part applies the arrival design to four measures of the manager's presence, built at the store-month level from the minute-level time-clock records. *Manager shifts* counts the shifts the general manager works in the month; *weekend shifts* and *night shifts* count the subset worked on a Saturday or Sunday and at night. *Co-presence* is the share of frontline worker shift-hours worked while the general manager is on the clock, computed by overlapping each worker's shift with the manager's daily presence interval. After a high-type arrival, the manager works more shifts, more of them on weekends and at night, and she spends more time on the floor with her workers.

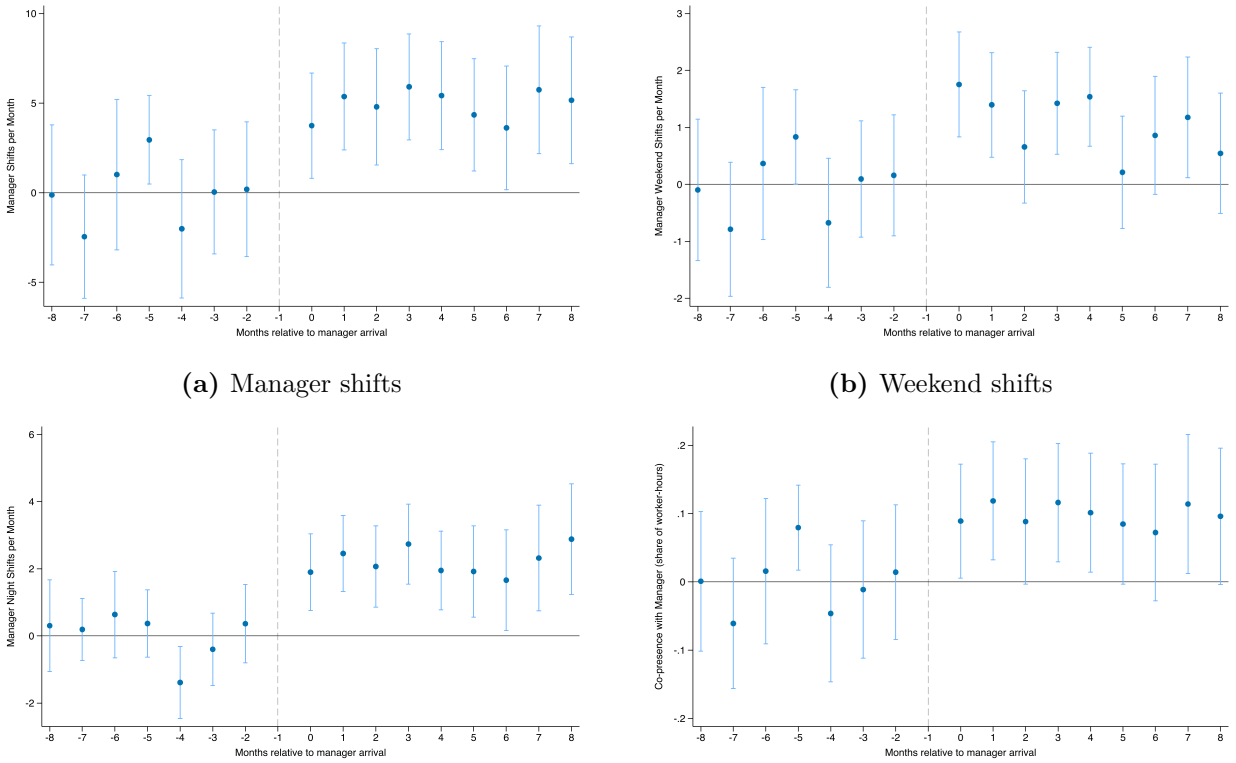
Table 6 reports the average effect over the nine months after each event, and Figure 3 shows its path month by month around the high-type arrival. The manager is there more. She works 1.5 more shifts per month on a base of 23.6, with weekend shifts up 0.39 on a base of 7.0, and night shifts up 0.66 on a base of 6.0, which translates to increases of five to eleven percent. Because she is present more while the store operates no differently, the frontline ends up being supervised more directly: co-presence rises 3.2 percentage points on a base near 57 percent. In the figure, all four measures rise in the month of the arrival and stay up through the eighth month, while the previous months scatter around zero. None of this happens when a low-type manager arrives: she moves none of the four measures. The

departure columns all shrink toward zero, consistent with the presence margin switching off when she goes. The added presence is broad rather than targeted at particular groups of workers (Appendix D.4).

Table 6: Managerial presence and manager–worker co-presence

	(1) Manager shifts/month	(2) Manager weekend shifts	(3) Manager night shifts	(4) Co-presence (worker-hrs)
HT manager arrival	1.5380***	0.3923**	0.6565**	0.0324***
	(0.4739)	(0.1583)	(0.2741)	(0.0121)
HT manager departure	−0.3205	−0.1342	−0.5222	−0.0238
	(1.6438)	(0.5279)	(0.6668)	(0.0385)
LT manager arrival	0.3250	0.1401	0.4689	0.0039
	(0.7416)	(0.2325)	(0.3773)	(0.0160)
LT manager departure	0.2035	−0.0832	0.3389	−0.0038
	(1.0316)	(0.3287)	(0.5016)	(0.0276)
Mean	23.58	6.99	6.04	0.575
Median	26.00	8.00	5.00	0.654

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a difference-in-differences regression of the store-month outcome on an indicator equal to one in the first nine months after the row's event (event months 0 to 8) and zero in every other month, with store, calendar-month, and year fixed effects and standard errors clustered at the store, on the stores that experience the row's event, estimated on store-month measures built from shift-level time-clock records. "Manager shifts" counts the general manager's worked shifts in the month; "Manager weekend (night) shifts" the subset on Saturday–Sunday (night hours). "Co-presence" is the mean share of a worker shift's hours overlapping the manager's daily presence interval. Mean and Median are of the dependent variable. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.



(a) Manager shifts

(b) Weekend shifts

(c) Night shifts

(d) Co-presence

Figure 3: The manager's presence around the high-type arrival

Notes: Each panel plots the coefficients from an event-study regression of the store-month outcome on event-month indicators, with store, calendar-month, and year fixed effects, around the arrival of a high-type manager for one measure of her own presence, all taken from the store's biometric time clock: (a) the shifts she works in the month; (b) her weekend shifts; (c) her night shifts; and (d) co-presence, the share of frontline worker-hours worked while she is also on the clock. The unit of observation is the store-month. The omitted period is −1. Store, calendar-month, and year fixed effects; ever-treated stores. The vertical line marks the handover month. Bands are 95 percent confidence intervals with standard errors clustered by store.

The manager then supplies the monitoring herself. The second part asks whether she also adds monitoring staff to the store. I apply the same arrival design to three measures of the store's monitoring apparatus: *loss-prevention staff*, the workers in the store's receiving, protection, and inventory-control positions per hundred workers; *supervisors*, area leaders, assistant managers, and managers as a share of the store's headcount; and *security staff*, the share of the headcount in security positions. Table 7 reports the results. After a high-type arrival, loss-prevention staff move by 0.25 per hundred workers on a base of 5.2, the supervisory share by −0.25 on a base of 25.7, and the security share by 0.08 on a base of 13.1. None is distinguishable from zero, and the intervals rule out the kind of staffing change that could explain a shrinkage reduction of this size. A low-type arrival does not move them either, apart from a small, marginally significant fall in loss-prevention staff.

Measured in shifts rather than headcount, the three margins do not move after a high-type arrival (Appendix D.5).

Presence is the empirical content this paper gives the monitoring term μ_m of the model: the manager makes the dismissal threat credible by being there to notice, in line with the evidence that monitoring deters opportunism (Nagin et al., 2002) and that workers lift effort when observed (Mas and Moretti, 2009).

Table 7: The store's monitoring apparatus on manager arrivals

	(1) Loss-prevention staff (per 100)	(2) Supervisors (% of staff)	(3) Security staff (% of staff)
HT manager arrival	0.2496	−0.2454	0.0750
	(0.1583)	(0.4051)	(0.1834)
LT manager arrival	−0.2071*	0.0391	0.1290
	(0.1117)	(0.4212)	(0.1333)
Mean	5.19	25.69	13.15
Median	4.71	25.00	12.96

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a difference-in-differences regression on an indicator equal to one in the first nine months after the row's event (event months 0 to 8) and zero in every other month, with the store-month outcome in the column: loss-prevention staff per hundred workers, and the shares of supervisory and security personnel in the store's headcount. Store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores. Mean and Median are of the dependent variable. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

5.3 The Team: Reorganized, Not Replaced

A good manager could also cut shrinkage by changing who works in the store—letting go of the workers who supply little care and replacing them with better ones. I test this with four measures of the team: turnover, the log of the month's separations per employee; upward moves, within-store reassignments to a higher tier; cross-area moves, reassignments across functional areas, both per hundred workers; and team size, the log of the store's headcount. The answer is that an HT manager reorganizes the team she inherits rather than replacing it. In her first months she moves existing workers up a tier and across areas. Separations rise only later—they are quits replaced one for one rather than dismissals, and headcount never changes. Because shrinkage falls in the arrival month, before turnover moves, the decrease does not come from dismissing workers.

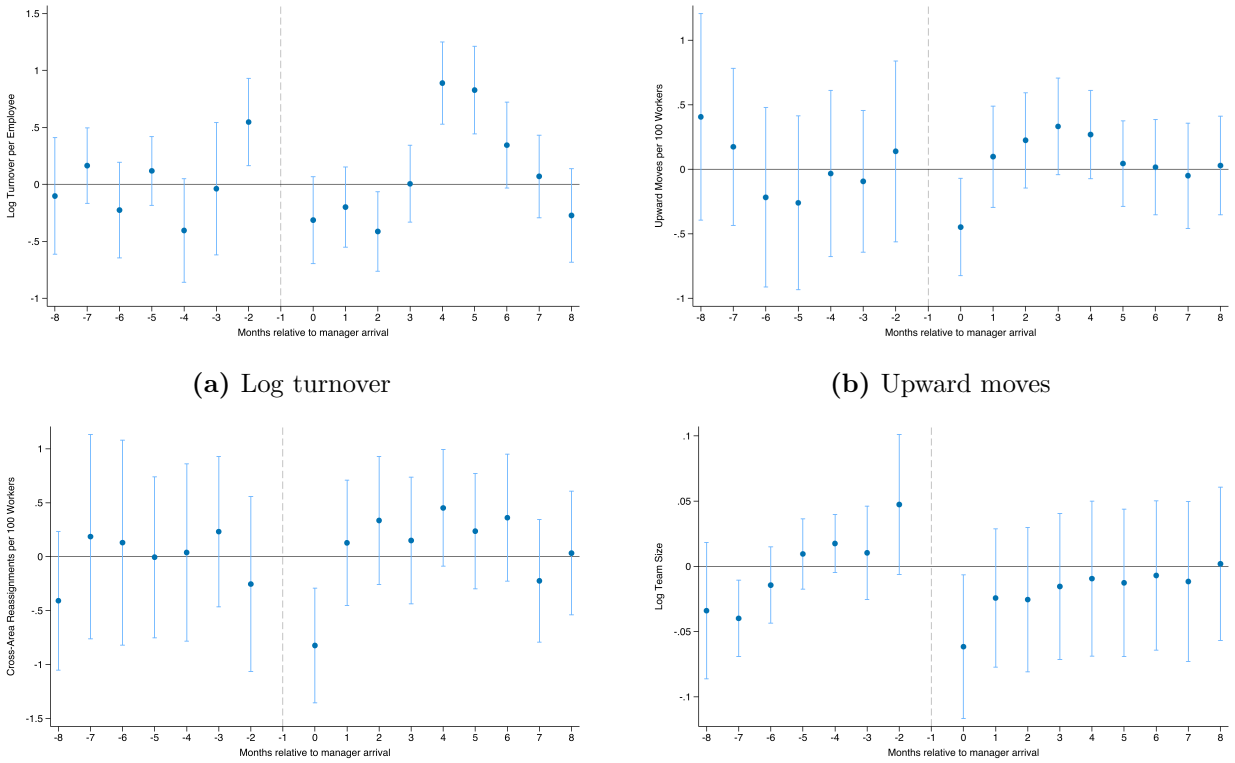
Table 8 summarizes the results, while Figure 4 shows the dynamics. Turnover rises 19 percent over the nine months after a high-type arrival, yet team size does not move. The

rise is quits (+0.17) and one-for-one replacement hires (+0.21), with no detectable rise in dismissals (+0.003, standard error 0.079; Appendix Table 18 reports the full decomposition). What she does with the workers she keeps is modest and internal: reassignments to a higher tier rise by 0.19 per hundred workers on a base of 1.0 and reassignments across functional areas by 0.21 on a base of 1.4, while the composition of the team—its diversity of age, tenure, education, gender, tier, and pay—remains unchanged (Appendix D). The cross-area moves concentrate in reassignments into client-facing roles, with moves into back-office roles, positive but imprecise, and no rise in moves into the loss-prevention chain (Appendix D.3); she redeploys the workers she has rather than staffing up inventory control. A low-type arrival moves none of these margins. Appendix D traces the other three transitions: a low-type arrival produces the same fourth-month increase in separations but only a modest, imprecise rise in reassignments, and after a low-type departure upward and cross-area reassignments increase.

Table 8: The team on manager arrivals and departures

	(1) Log turnover (p.e.)	(2) Upward moves (per 100)	(3) Cross-area moves (per 100)	(4) Log team size
HT manager arrival	0.1738**	0.1920***	0.2076**	0.0074
	(0.0724)	(0.0651)	(0.0920)	(0.0108)
HT manager departure	0.0519	0.1260	0.2021	−0.0081
	(0.1896)	(0.1608)	(0.1993)	(0.0248)
LT manager arrival	0.0245	−0.0048	0.0286	−0.0089
	(0.0610)	(0.0702)	(0.1134)	(0.0091)
LT manager departure	0.0279	0.3125**	0.5081***	0.0065
	(0.1486)	(0.1423)	(0.1896)	(0.0193)
Mean	−3.32	1.01	1.44	4.12
Median	−2.80	0.00	0.79	4.03

Each cell is the $\text{Post}_{[0,8]}$ coefficient from a difference-in-differences regression of the store-month outcome on an indicator equal to one in the first nine months after the row's event (event months 0 to 8) and zero in every other month, with store, calendar-month, and year fixed effects and standard errors clustered at the store, on the stores that experience the row's event. Turnover is the log of the monthly separation rate per employee; upward moves are within-store reassignments to a higher tier while cross-area moves are reassignments across functional areas, both per hundred workers. Mean and Median are of the dependent variable. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.



(a) Log turnover

(b) Upward moves

(c) Cross-area moves

(d) Log team size

Figure 4: The team around the high-type arrival

Notes: Each panel plots the coefficients from an event-study regression of the store-month outcome on event-month indicators, with store, calendar-month, and year fixed effects, around the arrival of a high-type manager for one measure of the store's team, from the personnel movement log: (a) log turnover, the monthly separation rate per employee; (b) upward moves, reassignments to a higher tier within the store, per hundred workers; (c) cross-area moves, reassignments across functional areas, per hundred workers; and (d) log team size. The unit of observation is the store-month. The omitted period is −1. Store, calendar-month, and year fixed effects; ever-treated stores. The vertical line marks the handover month. Bands are 95 percent confidence intervals with standard errors clustered by store.

6 Wage Premia and Shirking: Two-Way Fixed-Effects Evidence

The manager is one lever a firm holds over care; pay is another. A worker whose job pays more than she could earn elsewhere has something to lose if she is dismissed, and the threat of dismissal disciplines her care only to the extent of that rent. The model in Section 3 therefore predicts that shrinkage falls as the wage premium rises. This section tests that prediction.

Because the firm sets pay by position, the premium a worker earns moves only with the outside wage of her municipality. Outside wages vary widely across Mexico: the middle 90 percent of municipalities span about USD 150 to USD 440 a month in mean market wage (Appendix Figure 18), so the same position commands a very different premium depending on where the store is, and that premium moves over time as local wages move. I use this variation at two levels. At the store level, I compare stores that pay different premia in the same municipality and quarter; this is the level at which shrinkage is recorded, and the level of the paper's central outcome. At the worker level, I follow the same worker as the outside wage of her cell moves. The worker level answers a question that the store level cannot. A store that pays a higher premium may simply employ different workers—a premium can attract and retain more careful people—so a store-level gradient could reflect who works in the store rather than how a worker responds to her rent. The model's claim is about the second: the same worker supplies more care when she has more to lose. Following one worker, with worker and store fixed effects, removes every permanent attribute of the worker and of her store and isolates that response. Shrinkage is not recorded by worker, so this level is estimated on the complementary shirking margins—absence, turnover, shifts, and hours.

The store-level model is

$$\log Y_{it} \;=\; \delta \, \log W^{p}_{it} \;+\; Z_{it}\,\theta \;+\; \eta_{m(i)} \;+\; \tau_t \;+\; \nu_{it}, \tag{22}$$

where Y_{it} is inventory shrinkage of store i in quarter t; W^{p}_{it} is the store wage premium, the average of the worker premium in Equation 1 across the store's workers; $\eta_{m(i)}$ is a municipality fixed effect; τ_t is a year-by-quarter fixed effect; and Z_{it} optionally includes store format and log sales. Standard errors are clustered by municipality. The fixed effects absorb permanent differences across municipalities and shocks common to all stores in a quarter, so δ rests on movements in the local outside wage rather than on its level. Table 9 reports the estimates. The shrinkage gradient in column 1 is -0.113: a 10 percent lower premium is associated with about 1.2 percent more shrinkage. The gradient keeps its sign and significance when

I add store format and log sales and when I replace the year-by-quarter fixed effects with separate year and quarter effects (columns 2–4), and it is robust to normalizing shrinkage by sales, in value and in physical units (Appendix Table 27).

Table 9: Store-level TWFE: inventory shrinkage (Log–Log)

Wage premium (Log)	(1)	(2)	(3)	(4)
Coefficient	-0.113^{***}	-0.114^{***}	-0.045^{**}	-0.111^{***}
SE	(0.032)	(0.033)	(0.021)	(0.031)
Store format	No	Yes	Yes	No
Log sales	No	No	Yes	No
Municipality FE	Yes	Yes	Yes	Yes
Year×Qtr FE	Yes	Yes	Yes	No
Year + Quarter FE	No	No	No	Yes
Mean	12.41	12.41	12.41	12.41
Median	12.35	12.35	12.35	12.35

Store-quarter level. The dependent variable is the log value of inventory losses in Mexican pesos; Mean and Median are of the log outcome. Standard errors clustered at the municipality level. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

The complementary shirking margins move the same way (Appendix E). At the store level, a 10 percent lower premium is associated with about 1.4 percent more non-justified absenteeism, 1.7 percent more turnover, and 0.4 percent more total absenteeism, and with 0.3 percent fewer monthly shifts and 0.1 percent fewer daily hours (Appendix Table 28). Within the same worker, a larger premium goes with fewer absence days, total and non-justified, with lower odds of any absence, of any non-justified absence, and of separation, and with more monthly shifts and longer daily hours; every estimate keeps its sign and significance under controls for managerial tier, tenure, and age and under the choice of time fixed effects (Appendix Tables 30–31). For the margins observed at both levels, the signs match, and the magnitudes are close for absence days, shifts, and hours, while the non-justified absence gradient is larger at the store level than within the worker (Appendix Table 32).

6.1 Could the Outside Wage Move Shrinkage Directly?

The premium moves because the outside wage moves, so the gradient is the effect of the rent only if the outside wage does not reach shrinkage by another road. I consider six.

The incentive to take merchandise from outside. Part of what goes missing from a store is taken by people who do not work there, and their incentive depends on what they can earn legally; where local wages rise, the return to legal work rises and property crime falls (Gould et al., 2002; Machin and Meghir, 2004). This channel works against the estimate.

Shrinkage is written as the sum of a part that falls with the premium and a part that falls with the local wage. A higher outside wage lowers the premium, which raises the first part, and lowers the second. The estimated gradient is the rent effect net of the outside-theft effect, so it understates the rent effect in absolute value.

Local crime. A local boom could still move crime for other reasons. It does not move with the premium: the premium does not predict commercial robbery, homicide, or the share of residents who report feeling unsafe in the store's municipality, and the shrinkage gradient is unchanged when I control for them (-0.114 with the crime measures, -0.139 adding perceived insecurity, against -0.113 in the baseline).

Local unemployment. A slack local labor market lowers what a worker can expect outside and could change who shops and who works in the store. Controlling for the unemployment rate of the store's local labor market leaves the gradient at -0.120.

Local prices. Shrinkage is recorded in Mexican pesos, so a local wage increase that raises local prices would raise the value of the same missing merchandise. The gradient is not a price effect: measured in physical units it is larger, -0.176, and it is -0.146 for units lost per unit sold.

Product mix. Where local incomes are higher a store may carry more perishable merchandise, which is lost more easily. The premium does not predict the perishable share of the store's inventory.

Sales. A local boom raises sales, and a store that sells more has more to lose. The gradient survives the control: it is -0.069 conditional on log sales and -0.045 adding store format (Table 9, column 3), and -0.062 for shrinkage as a share of sales (Appendix Table 27).

7 Managers and the Wage Premium: Complements in Discipline

Care is attention a firm cannot measure and a supervisor observes only imperfectly. It cannot be traced to any particular worker. What the firm records is a loss traced to a unit; a store, a restaurant, a plant. The monitor is therefore indispensable; unlike absence, punctuality, or the length of a break, care leaves no record unless someone is there to observe it. The worker, in turn, cannot be paid for it directly; there is no pay for performance on an outcome that cannot be traced to her, so the premium the firm pays over the local market is the incentive it has left. Sections 5 and 6 show that each lever works: shrinkage falls when a better manager arrives, and it falls where the premium is higher. The question is whether each can sustain care independently or whether the two are more effective when deployed jointly.

The question matters because the two levers do not cost the same. In the average store, with about 67 workers and a premium of USD 93 a month each, raising the premium by 10 percent costs about USD 625 a month. The gradient of Section 6 associates that raise with 1.1 percent less shrinkage; about USD 180 a month. On the shrinkage margin alone, a dollar of premium returns about thirty cents. A high-type manager, in contrast, costs the same as a low-type one, because pay is fixed by position; the cost of better management is the cost of developing managers, which at least theoretically is lower than the cost of raising the wage of every worker in the store. If the two levers are substitutes, a firm can economize on the expensive one. If they are complements, it cannot; the manager needs a rent to threaten, and the rent needs a monitor.

This section estimates the arrival design of Equation (21) separately in high- and low-premium environments. The arrival of a high-type manager reduces shrinkage by 11.8 percent where the premium is high and by 4.2 percent where it is low; relative to a low-premium store, the same arrival lowers shrinkage by a further 8.1 percent. In the average store, that difference is about USD 1,400 of merchandise a month. To obtain the same additional reduction through pay alone, the gradient of Section 6 implies a premium about 110 percent higher, more than double; about USD 6,900 a month in the average store.

7.1 Design

I take the high-type arrival design of Section 5 and ask one question of it: is the shrinkage reduction a high-type manager delivers larger where the store's wage premium is high?

I classify each store's municipality, in the month its manager arrives, as a high-wage-premium (HWP) or low-wage-premium (LWP) environment: HWP if the premium the firm

pays over the local ENOE market wage lies above that municipality's own median over the sample, LWP otherwise. Fixing the environment at arrival makes it pre-determined with respect to everything that follows and constant within a store's event window. I then re-estimate the arrival difference-in-differences of Equation (21), interacting the $\text{Post}_{[0,8]}$ indicator with this environment, separately for high- and low-type arrivals.

I include one alternative design. It pools the high- and low-type arrivals in a single regression and interacts the post-arrival indicator with the type of the arriving manager and with the wage environment,

$$\begin{aligned} Y_{jt} \;&= \beta_1 \,\text{Post}_{jt} + \beta_2 \,\text{Post}_{jt} \times \text{HT}_j + \beta_3 \,\text{Post}_{jt} \times \text{HWP}_j \\ &\quad + \beta_4 \,\text{Post}_{jt} \times \text{HT}_j \times \text{HWP}_j + \varphi_j + \lambda_{m(t)} + \xi_{y(t)} + \varepsilon_{jt}, \end{aligned} \tag{23}$$

with the fixed effects and clustering of Equation (21). The coefficient β_4 is the triple difference: how much more the distance between a high-type and a low-type arrival widens where the premium is high. The effect in each of the four cells is a linear combination of the coefficients.

7.2 Good Managers Accomplish More Where the Premium Is High

The complementarity is the difference between the two environments in what a high-type arrival accomplishes, and Table 10 reports this in the upper panel. A high-type arrival reduces shrinkage by 11.8 percent where the premium is high and by 4.2 percent where it is low, so the same manager lowers shrinkage by a further 8.1 percent, nearly three times the low-premium effect, where the premium gives her dismissal threat something to confiscate; the difference is significant at five percent. The lower panel repeats the exercise for low-type arrivals. Shrinkage rises by 6.1 percent where the premium is low and by 3.8 percent where it is high; the second estimate, and the difference of 2.2 percent between the two, are not distinguishable from zero. A weak monitor is bad for the store, whatever it pays: with little detection, the rent has nothing to threaten on the care margin, so a high premium does not replace the manager.

The result does not depend on how shrinkage is measured or on how the environment is defined. Appendix H repeats the exercise with log shrinkage over sales and log shrinkage per worker: the difference between environments is 5.9 and 9.6 percent, significant at ten and at one percent. Under three alternative definitions of the wage environment, the sign of the difference and the ordering of the two environments are preserved (Appendix Table 42 and Appendix Figure 21).

Table 10: Manager arrivals by wage environment at arrival

	Log shrinkage value
HT arrival, high-premium (HWP)	-0.126^{***}
	(0.032)
HT arrival, low-premium (LWP)	-0.043^{**}
	(0.018)
Difference (HWP − LWP)	-0.084^{**}
	(0.037)
LT arrival, high-premium (HWP)	0.037
	(0.033)
LT arrival, low-premium (LWP)	0.059^{***}
	(0.021)
Difference (HWP − LWP)	−0.022
	(0.039)
Mean	12.43
Median	12.35

$\text{Post}_{[0,8]} = 1$ for event times 0 through 8, interacted with the wage environment of the store's municipality in the arrival month (HWP = premium above the municipality's median; LWP otherwise). Each panel reports the arrival effect in each environment and then their difference (the complementarity); high-type arrivals above the rule, low-type below. Store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores only. A manager is high-type if her fixed effect for log shrinkage value lies below the median of the manager-effect distribution within her connected set, and low-type otherwise; an arrival is the first time a store comes under a manager of the given type for two consecutive months, and the first of those months is the event date; the outcome is the log value of inventory losses. Mean and Median are of the log outcome. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

Figure 5 traces the monthly path behind Table 10. Shrinkage falls in both environments when a high-type manager arrives, and from the second month after the arrival it falls by more where the premium is high. After a departure the two environments do not differ.

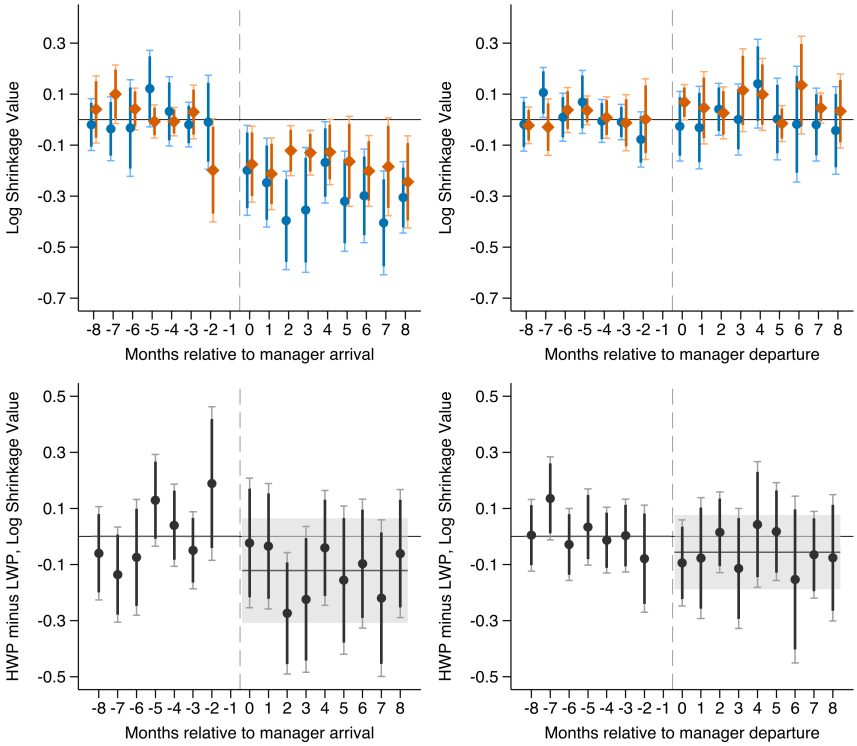
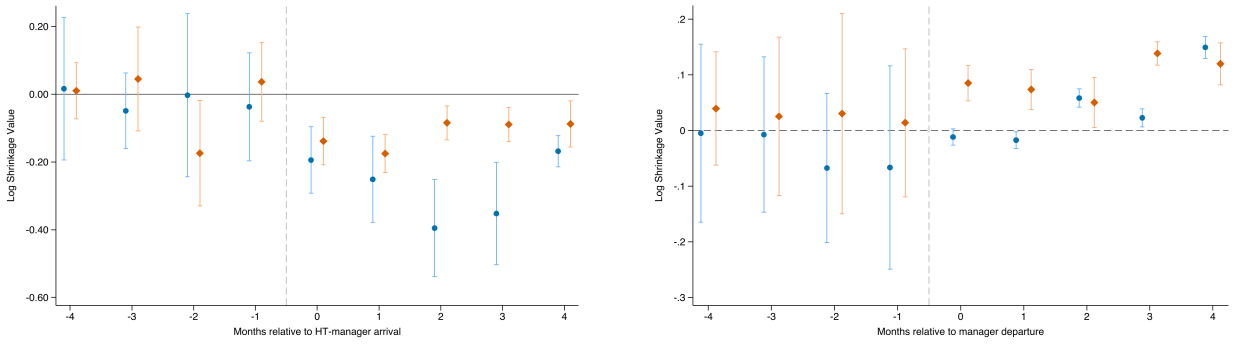


Figure 5: Log shrinkage value around high-type transitions, by wage environment

Notes: Event-study coefficients of log shrinkage value around the arrival (panel a) and the departure (panel b) of a high-type manager, estimated separately for high-premium stores (HWP, blue circles) and low-premium stores (LWP, orange diamonds); panels (c) and (d) plot the HWP minus LWP difference in each month. The wage environment is that of the store's municipality two months before the event: HWP if its premium is above the municipality's median, LWP otherwise. Coefficients are relative to the average of the eight months before the event; the coefficient of the handover month, the month before the event, is estimated but not plotted. Store, calendar-month, and year fixed effects. The vertical line separates the months before the event from the event month and the months after it. Thin whiskers are 95 percent confidence intervals and thick segments are 90 percent intervals, with standard errors clustered by store. In panels (c) and (d) the horizontal line and the shaded band are the average difference over the event month and the eight months after it, with its 95 percent interval.

The arrivals are staggered, so stores treated earlier serve as controls for stores treated later. Figure 6 re-estimates the arrival dynamics with the imputation estimator of Borusyak et al. (2024), which builds each treated store's counterfactual from never-treated stores and from its own pre-arrival months only. The pattern is the same: shrinkage falls in both environments, and by more where the premium is high. The two paths coincide on impact; the high-premium path separates in the second and third months, where the two intervals do not overlap, and the two converge afterward. The complementarity also survives alternative ways of labeling managers. Rebuilding the label from the manager's fixed effect on non-justified absence, an outcome the time clock records, gives a differential of -5.8 percent,

significant at five percent. Permuting residuals within store and re-running the entire pipeline one thousand times places the observed differential outside the full placebo distribution; net of the mechanical component it is −6.0 percent. Estimating the label on odd months and testing it on even months gives −6.6 to −8.1 percent (Appendices I and J).



(a) Arrival of a high-type manager

(b) Departure of a high-type manager

Figure 6: Log shrinkage value around high-type transitions, by wage environment: imputation estimator

Notes: Imputation estimator of Borusyak et al. (2024) for log shrinkage value around the arrival (a) and departure (b) of a high-type manager, estimated separately for high-premium (HWP, blue circles) and low-premium (LWP, orange diamonds) stores. The counterfactual for each treated store-month is built from never-treated stores and from the store's own months before the event, so no already-treated store serves as a control and there is no omitted reference period: post-event points are deviations of observed from imputed shrinkage, and pre-event points are placebo coefficients estimated relative to months more than four months before the event. Only events whose pre-period is observed enter, far fewer than the events behind the pooled difference-in-differences estimates reported in the text. The unit of observation is the store-month. Whiskers are 95 percent confidence intervals with standard errors clustered by store; points are offset horizontally so the two series do not overlap.

Table 10 estimates high- and low-type arrivals in separate regressions, so each panel compares a store with itself and with the other stores of the same type. Figure 7 puts the four combinations of manager type and wage environment into the single regression of Equation (23). Estimated together, the four cells share the same controls and the same fixed effects, so they can be read against one another, and the gap between them is a coefficient the regression tests directly, the triple difference β_4. The sample is also stricter: only arrivals whose months before the event are observed enter, with the stores without such an arrival as controls, so every cell compares a store before and after a manager it is seen to receive.

Figure 7 sets the firm's two decisions side by side: whom it puts in charge of a store, and how much it pays its workers above the local market. A high-type manager lowers shrinkage wherever she arrives—by about 7.1 percent in a low-premium store and by about 18.5 percent in a high-premium one—so the same manager recovers about two and a half times as much where workers have more to lose. A low-type manager raises shrinkage

wherever she arrives—by about 10.6 percent where the premium is low and by about 5.4 percent where it is high—so a high premium does not turn the arrival of a weak monitor into a gain. All four effects are distinguishable from zero. The distance between the two types of manager is wider in the high-premium environment, the direction complementarity predicts, but high-premium arrivals are scarce and the figure does not measure that widening with precision; the powered test remains the per-type interaction of Table 10.

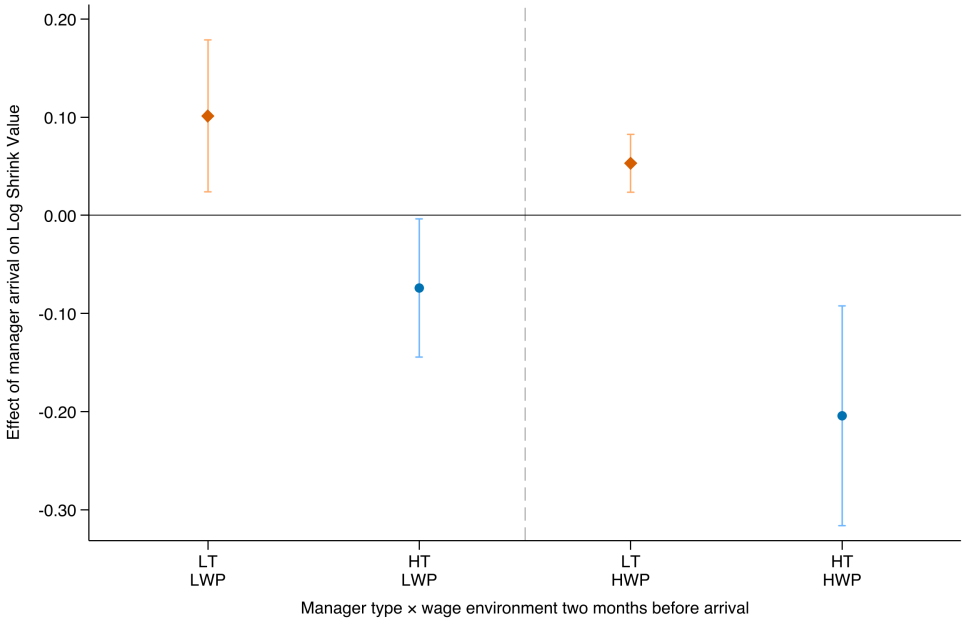


Figure 7: Effect of a manager arrival on log shrinkage value, by manager type and wage environment

Notes: Each point is the effect of a manager arrival on log shrinkage value in one of four cells, computed as a linear combination of the coefficients of a single pooled regression that interacts the post-arrival indicator with the type of the arriving manager and with the store's wage environment: store, calendar-month, and year fixed effects. A manager is high-type (HT) if her fixed effect in the decomposition of log shrinkage value lies below the median of the connected set, so that her stores lose less merchandise under her than under the median manager, and low-type (LT) otherwise; the wage environment is high-premium (HWP) if the premium of the store's municipality is above the municipality's median and low-premium (LWP) otherwise. Whiskers are 95 percent confidence intervals with standard errors clustered by store; the dashed vertical line separates the two environments.

7.3 Does the Manager Do More Where the Premium Is High?

The complementarity admits two readings, which differ in what they imply for the manager's conduct: a good manager could supply more discipline where the rent is high, or supply the same actions everywhere and have each action return more where the rent is high. To separate them, I re-estimate the arrival design with each action of Section 5 as the outcome, the presence and team margins as defined in that section, and ask whether the arrival changes

the action by more where the premium is high. For a store-month action a_{jt} the regression is the difference-in-differences companion of Equation (21), with the post-arrival indicator interacted with the wage environment at arrival as in Table 10,

$$a_{jt} \;=\; \beta\,\mathrm{Post}_{jt} + \gamma\,\mathrm{Post}_{jt} \times \mathrm{HWP}_{j} + \varphi_{j} + \lambda_{m(t)} + \xi_{y(t)} + \varepsilon_{jt}, \tag{24}$$

where β is the arrival effect on the action in a low-premium store, $\beta + \gamma$ the effect in a high-premium store, and γ the differential between the two. Figure 8 plots γ for each action, divided by the standard deviation of the action so that margins measured in different units can be read on one axis; Table 11 reports the same estimates margin by margin, in the units of each action.

The pattern in the manager's actions is a tendency rather than a finding. Her shifts per month, her weekend shifts, and her co-presence with the workers' hours all rise by more in high-premium stores, but only the differential in shifts is significant, at ten percent, and the others are not; the data cannot say whether the manager is more present where the premium is high. Turnover shows the same tilt. It rises by 36 percent where the premium is high and by 12 percent where it is low, and the rise is in quits, which rise by 36 percent in high-premium stores and do not move detectably in low-premium ones, while dismissals do not move in either environment (Appendix Table 37). The differentials in turnover and in quits are not distinguishable from zero, so the comparison across environments is again a tendency. The one margin on which the two environments differ with precision is enforcement. Suspensions rise by 0.31 per hundred workers after a high-type arrival in a high-premium store and fall by 0.14 in a low-premium one; a differential of 0.45, significant at five percent. This is the mechanism of Proposition 4: the sanction is the manager's own act and follows the rent, so where the premium gives her threat something to confiscate she uses it more. The team is not recomposed: team size, upward and cross-area moves, and the gender, occupational, and diversity mix of the store respond no differently across environments, with the supervisory share as the one exception among the manager's inputs (Appendix G).

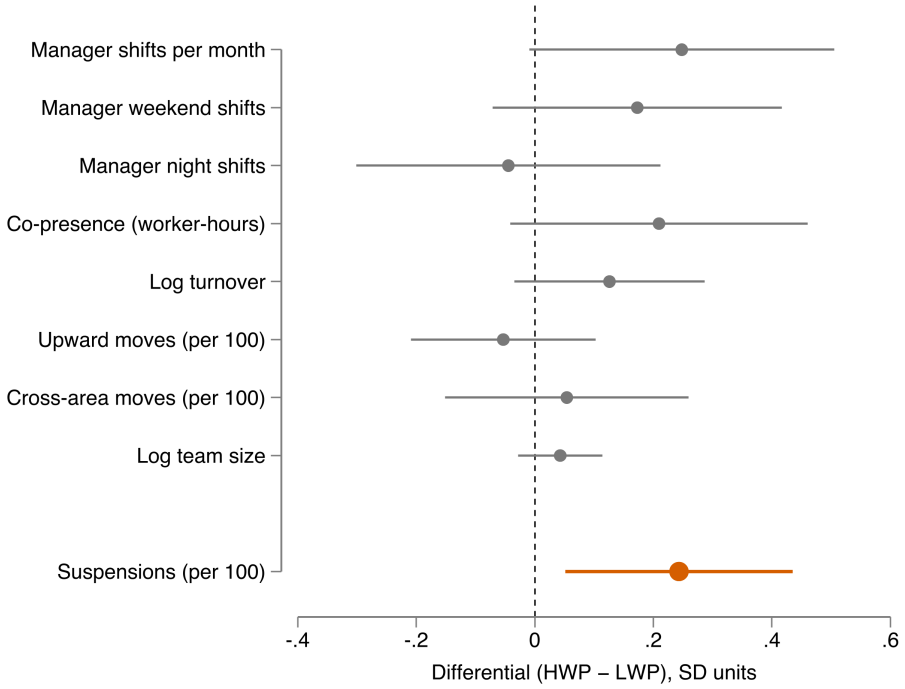


Figure 8: The differential between wage environments in each of the manager's actions after a high-type arrival

Notes: Each point is the difference between high- and low-premium stores in the effect of a high-type manager's arrival on one of her actions, the coefficient γ of Equation (24), divided by the standard deviation of the action so that the margins are comparable; a point to the right of zero means the arrival moves the action by more where the premium is high. A manager is high-type if her fixed effect in the decomposition of log shrinkage value lies below the median of her connected set, so her stores lose less merchandise under her than under the median manager; the wage environment is high-premium (HWP) if the premium of the store's municipality in the arrival month is above the municipality's median and low-premium (LWP) otherwise. Presence margins: shifts per month, weekend shifts, night shifts, and co-presence with frontline worker-hours. Team margins: log turnover, upward and cross-area moves per hundred workers, and log team size. Enforcement: suspensions per hundred workers, in orange. Store, calendar-month, and year fixed effects; standard errors clustered by store; ever-treated stores. Bars are 95 percent confidence intervals. The figure shows which actions differ between environments, not the size of the manager's effect on any of them, which Table 11 reports.

Table 11: Does the manager do more where the premium is high? Every margin crossed with the wage environment at arrival

	Effect in LWP stores	Effect in HWP stores	Difference (HWP − LWP)
Panel A. The manager's actions			
Manager shifts per month	0.916	3.042***	2.126*
Manager weekend shifts	0.252	0.745***	0.493
Manager night shifts	0.715**	0.497	−0.218
Co-presence (worker-hrs)	0.020	0.063***	0.043
Log turnover (p.e.)	0.115	0.309***	0.194
Upward moves (per 100)	0.208**	0.131	−0.077
Cross-area moves (per 100)	0.173	0.272*	0.098
Log team size	0.001	0.023	0.023
Panel B. Enforcement and the return			
Suspensions (per 100)	−0.140	0.311**	0.451**
Log shrinkage value	−0.043**	−0.126***	−0.084**

Each row is a difference-in-differences regression of the indicated store-month outcome on a post-arrival indicator equal to one in the first nine months after the high-type arrival (event months 0 to 8) and zero in every other month, interacted with the wage environment fixed at the month of the high-type arrival (the municipality's premium above its own median): store, calendar-month, and year fixed effects; standard errors clustered at the store; ever-treated stores. Columns: the arrival effect in low- and high-premium stores and their difference. Rates are per hundred workers. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

What the subsection establishes is narrower than the two readings it opened with, and sharper. The manager may react to the rent with her presence: the tilt appears in every presence margin, but the differences are too imprecise to say that she does, and I do not claim this. What the data do say is that she enforces more where the premium is high, and that quits rise where she arrives at a high-premium store and not detectably where the premium is low. Both run through the rent. The suspension confiscates part of it, so it has bite only where there is something to confiscate; and a worker who will not supply the care demanded by a stricter monitor has, where the premium is high, a job worth keeping only by complying, so the arrival sorts some of them out. The complementarity in shrinkage is therefore the product described by the model, a monitor whose threat is worth more where the rent is higher, and not, as far as the data can tell, a manager who works harder where the firm pays more.

8 Minimum-Wage Hikes at the Border: Moving the Premium by Statute

Section 6 found that shrinkage falls where the wage premium is higher: a premium 10 percent lower is associated with about 1.2 percent more shrinkage. Section 7 found that the premium works through the manager, and accomplishes most where a good manager is in charge. Both results rest on movements in the local outside wage, which the firm does not control but which local conditions can move together with shrinkage. This section exploits the January minimum-wage hikes, which move the premium the firm pays over the legal floor by statute, and by more in the border zone than in the interior. The design adds to the evidence so far a shock whose size and date are set nationally, for reasons unrelated to any store. I then ask whether the manager on the floor changes the damage done by the hike. The model of Section 3 predicts that a small loss of rent moves care only where monitoring is weak: in a store whose manager already holds her workers at full compliance, the squeeze should leave shrinkage unchanged (Proposition 5).

Mexico's minimum-wage policy offers a clean way to move the wage premium by statute. The country keeps two statutory floors—a general one and a higher one in the northern border zone—and each January both rise by the same percentage, set by a federal commission and published in advance. The hikes move the premium in a way not chosen by the firm. Because the same percentage applies to a higher base, the step in Mexican pesos is larger at the border, so a single date moves border and interior stores by different amounts. Because the firm sets pay centrally by position and does not move its scale with the floor, the hike changes the premium its workers earn over the legal floor rather than a wage decided by the firm. And because the firm operates stores on both sides of the zone boundary, observed month by month, the same firm, under the same policies, is seen under two different shocks.

The exercise complements the two-way fixed-effects estimates of Section 6 in its source of variation and in the threat it removes. The gradient of Equation (22) is identified from movements in the ENOE outside wage of a store's municipality, net of municipality and year-by-quarter fixed effects. Its remaining threat is a local shock, a change in local demand or in local labor-market conditions, which moves outside wages and shrinkage together. The minimum-wage design moves a different component of the premium, the floor rather than the market wage, and moves it by statute; the size and the date of each hike are set nationally, for reasons unrelated to any store. The shock is therefore unrelated to local conditions by construction, and it is dated to a single month, so the months before it test whether border and interior stores were already diverging. What the design gives up is generality: it estimates the response to one kind of premium movement, in one part of the firm.

The exercise is an event study around each January hike. The step in Mexican pesos is about 1.5 times larger at the border and the firm's scale is the same on both sides, so the border store's premium over its floor compresses by more than the interior store's. The border stores are the treated group, the interior stores of the same format are the control group, and the event is the January in which the new floor takes effect. I follow both groups from six months before to six months after each hike, and ask whether border shrinkage rises relative to interior shrinkage once the floor moves.

The border floor is recent. The government created the *Zona Libre de la Frontera Norte* in January 2019, and since 2020 the two floors have risen by the same percentage every January, which holds their ratio near 1.51. The hikes of January 2023 and January 2024 raised both floors by twenty percent, from permanently different bases (CONASAMI). About a quarter of the firm's stores lie inside the border zone.

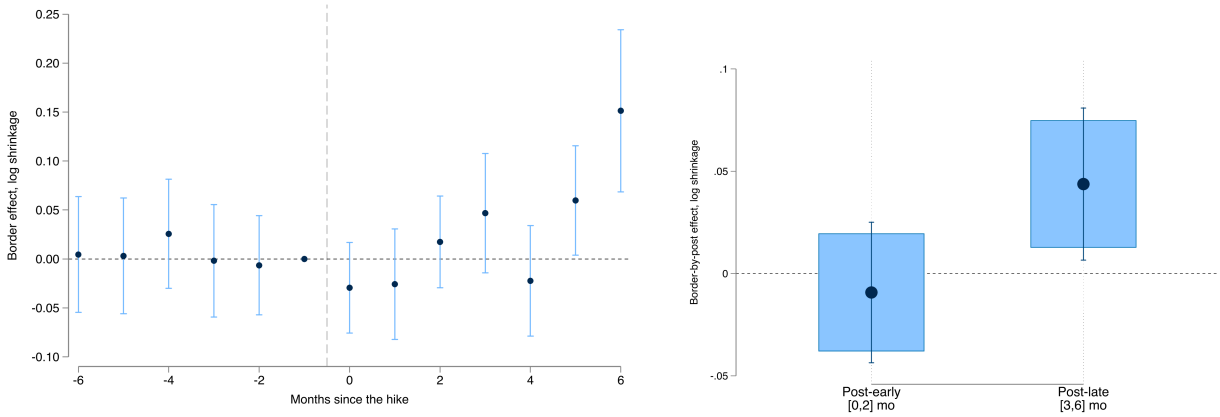
I estimate, for store k of format f in hike h at event month τ,

$$Y_{kh\tau} \;=\; \alpha_{k,h} \;+\; \gamma_{f(k),m} \;+\; \sum_{s \neq -1} \beta_s \left(\text{Border}_k \times \mathbf{1}[\tau = s] \right) \;+\; \varepsilon_{kh\tau}, \tag{25}$$

where $Y_{kh\tau}$ is log shrinkage net of sales and the two hikes are stacked as separate experiments (Cengiz et al., 2019). The store-by-hike fixed effect $\alpha_{k,h}$ compares each store with itself within each hike, so no border–interior level difference enters the estimate. The format-by-calendar-month fixed effect $\gamma_{f(k),m}$ absorbs the seasonal pattern of each store format, so a border store is set against interior stores of its own format in the same month of the year. Border_k marks stores inside the border zone, $\mathbf{1}[\tau = s]$ marks event month s, and the month before the hike is the omitted category, so each β_s is the border–interior gap in month s relative to December. Standard errors are clustered by store. The summary replaces the event-month indicators with two, $\text{PostEarly}_{[0,2]}$ and $\text{PostLate}_{[3,6]}$, for months zero to two and three to six after the hike, each measured against the two months before it. The comparison follows the border-comparison tradition of the minimum-wage literature (Dube et al., 2010).

If the premium disciplines, a statute that compresses it should raise shrinkage. Figure 9 tests this. Panel (a) follows the border–interior difference in shrinkage month by month; panel (b) averages it over post-early, months zero to two after the hike, and post-late, months three to six. Post-early, border shrinkage is unchanged relative to the interior (-0.009, $p = 0.60$). Post-late, it is 4.5 percent higher ($p = 0.022$). The timing is informative. The response is dated to the months in which the smaller premium persists, not to the day the law changes, as expected if the effect operates through the rent; the worker's incentive to exert care falls as the value of keeping the job falls. Section 8.1 splits this response by the manager on the

same clock.[10]



(a) Event study. **(b)** Post-early and post-late.

Figure 9: Shrinkage in border stores around the minimum-wage hikes

Notes: Panel (a) plots the border × event-month coefficients of a stacked event study of the two January hikes, with store-by-hike and format-by-calendar-month fixed effects, on log shrinkage net of sales: each point is the difference between border and interior stores in that month relative to the month prior to the January hike, which is the omitted category; the two January hikes in the sample are stacked as separate experiments and the vertical line marks the hike. Panel (b) plots the same difference averaged over post-early, months 0 to 2 after the hike, and post-late, months 3 to 6, each measured against the two months before the hike. Border stores are those in the zone with the higher statutory minimum. The unit of observation is the store-month; store-by-hike and format-by-calendar-month fixed effects; standard errors clustered by store. Whiskers are 95 percent confidence intervals; in panel (b) the box is a 90 percent interval.

The natural next test is the first stage: did the hike lower the wage premium as the paper measures it, the firm wage over the ENOE outside wage? That event study is underpowered. The outside wage is observed by quarter and the panel covers two and a half years, so the six months on each side of a hike contain two quarters, too few to trace the premium around the event. I instead show the part of the first stage that the firm's own records measure month by month: the worker's premium over her statutory floor. Figure 10 is the event study of Equation (25) at the worker level for the first of the two hikes, with that premium, in Mexican pesos per day, as the outcome, a worker fixed effect in place of the store effect, and a position-by-format-by-month fixed effect, so a border worker is compared with an interior worker in the same position and month. Both terms of the outcome are known exactly, the contractual wage from payroll and the floor from the statute, so it moves only when the floor or the firm's scale moves. The dashed line is the full statutory squeeze, the fall that would occur if the firm's wages did not move at all. The border worker's premium over her floor

[10]Post-late, the monthly path is uneven and climbs toward the end of the window, so the average over the four months is what should be read, not any single month; the design dates the response to the post-late months as a whole, not to a particular month.

drops in the month the new floor takes effect and stays down for the rest of the window, and the drop is about half the statutory squeeze. The premium therefore fell and remained lower, but by less than the statute implies: the 4.5 percent of Figure 9 is a response to the premium workers actually experienced, half the size of the one the statute implies.

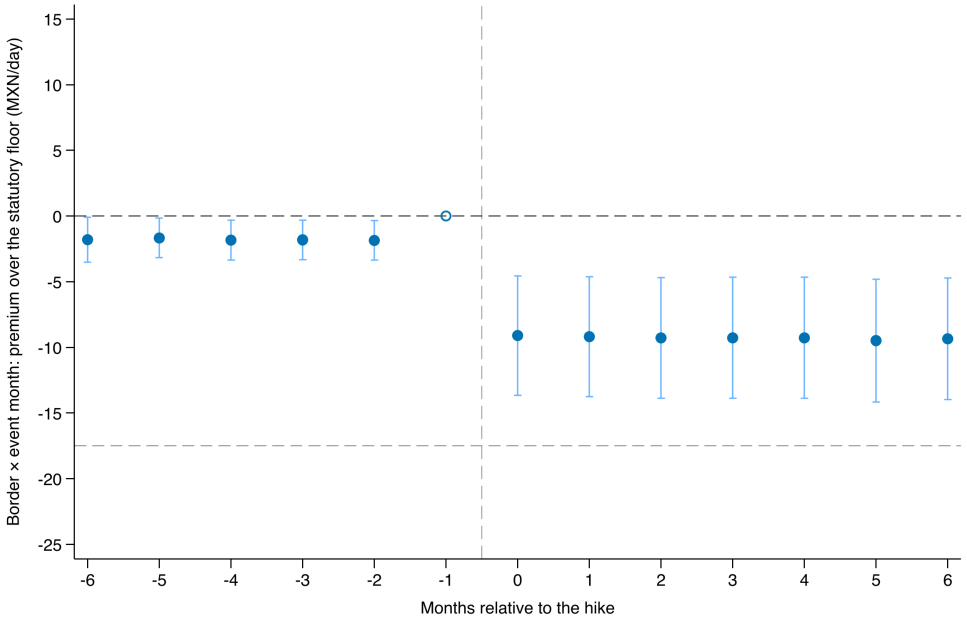


Figure 10: The worker's premium over the statutory floor around the first January hike

Notes: Border × event-month coefficients on the worker's contractual daily wage minus the statutory minimum of her zone, in pesos per day, around the first of the two January hikes. The sample is non-managerial full-time workers present throughout the window; the unit of observation is the worker-month. The month before the hike is the omitted category and is drawn as a hollow marker. Worker fixed effects and position-by-format-by-calendar-month fixed effects; standard errors clustered by store; whiskers are 95 percent confidence intervals. The dashed horizontal line is the full statutory squeeze: the difference between the border and interior floors' steps, in pesos per day.

The question of whether the damage depends on the manager on the floor when the shock lands is taken up in Section 8.1.

8.1 The Monitor at the Moment of the Shock

This subsection brings the manager into the border design. Section 7 showed that a high-type manager reduces shrinkage by more where the premium is high, and Figure 9 showed that the hikes raise border shrinkage in the months after the hike. The question here is which border stores bear that rise: every store, or the stores where the manager is seldom on the floor? The level result does not settle it. In levels, a monitor's threat is worth more where the rent is larger. A small cut in the rent is a different object. Where the manager is on the

floor and her workers already supply full care, a slightly smaller premium can still leave the job worth keeping by complying, so care need not move; where she is seldom present, care sits below that ceiling and responds to any change in what the job is worth. The model of Section 3 allows either pattern, so which one holds is an empirical question.

The design is the stacked border event study of Equation (25), summarized by the post-early and post-late indicators. I split the border stores into two groups by the general manager's floor presence, measured from the biometric time clock in the December before each hike and divided at the median within format-by-hike cells, and interact each post indicator with the two groups. Both groups are compared with the same interior stores, so each group's estimate is its own border–interior difference, and the object of interest is the difference between the two post-late responses. Because presence is measured before the hike, the hike cannot move the split. The cost is precision: each group holds half of the border stores, so the difference between the groups is estimated less precisely than the pooled effect of Figure 9.

Figure 11 shows the results. The two groups differ in both periods. Where the manager is on the floor, shrinkage falls post-early, by 4.8 percent relative to sales ($p = 0.10$), and the post-late months show no detectable effect (+1.9 percent); the erosion of the premium is, in effect, insured by the monitor. Where she is not, there is no post-early effect, and post-late shrinkage rises by 5.0 percent ($p = 0.02$). The level complementarity of Section 7 and this pattern are two readings of the same deterrent: a present manager makes the rent worth more, and keeps her workers far enough above the point of indifference that a small loss of rent does not move them.

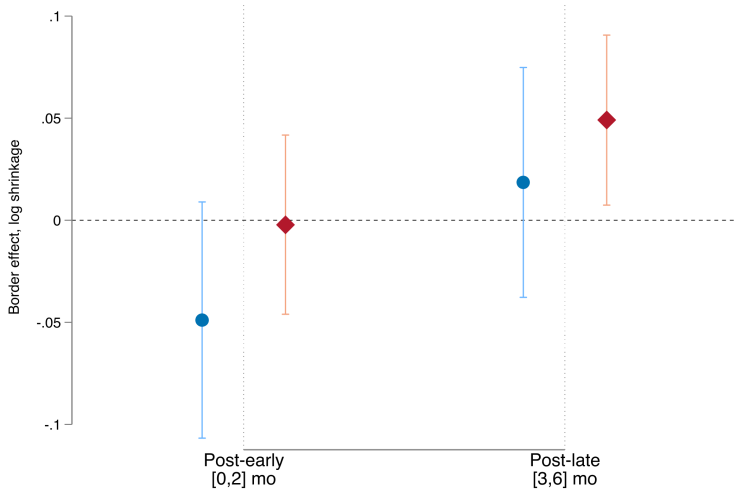


Figure 11: Shrinkage in border stores around the hikes, by the manager's floor presence

Notes: Border × phase coefficients on log shrinkage net of sales, both January hikes stacked, from a stacked border-versus-interior regression with store-by-hike and format-by-calendar-month fixed effects, the post-hike months grouped into post-early and post-late periods, for two groups of border stores: those whose general manager is on the floor often (blue circles) and those whose manager is seldom there (red diamonds). Presence is measured from the store's biometric time clock; the split is at the median presence within format-by-hike cells, on the manager in place the December before each hike. The horizontal axis has two periods, both measured against the two months before the hike: post-early, months 0 to 2 after the hike, and post-late, months 3 to 6. Store-by-hike and format-by-calendar-month fixed effects; whiskers are 95 percent confidence intervals, standard errors clustered by store.

9 Conclusion

In many jobs, effort takes the form of care: attention a firm cannot measure and a supervisor observes only imperfectly, since a lapse is caught only when someone is watching. The barista who gets an order right, the cook who keeps portions to the recipe, the line worker who lets no defective unit through, the clerk who keeps merchandise from going missing—each supplies care. What the firm sees is the residue: an order remade, food thrown out, a part scrapped, a gap on the inventory count; a cost it can record, but not trace to any particular worker. That is the firm's problem with care. Because the loss cannot be traced to a particular worker, measures to control it are difficult to implement; a firm cannot do pay for performance for care it does not observe, nor charge the loss to whoever caused it. Since the residue is scattered across many small failures, it is rarely recorded systematically at all; where it is, it is large. In United States retail, inventory shrinkage—merchandise that goes missing from stores—reached $112 billion in 2022, about 1.6 percent of sales, on the order of grocers' net profit margin (National Retail Federation, 2023; Damodaran, 2026); the most recent global survey puts it at a similar 1.8 percent of sales worldwide (Tyco Retail

Solutions and PlanetRetail RNG, 2018). Retailers treat it as a security problem, where they lock merchandise behind glass, expand video surveillance, and increase loss-prevention budgets, while some chains have closed stores because of it (Selyukh, 2024). Despite the cost of lapses in care for firms, it remains unclear which levers a firm holds to sustain care, and there is no systematic way to analyze the problem that could guide a firm in choosing among them.

I study inventory shrinkage as a measure of care and analyze two of the levers a firm holds to sustain it: monitoring—through its managers—and the wage premium. Using manager rotations that are unrelated to store shrinkage, I estimate each manager's effect on shrinkage and classify managers as high-type or low-type by their fixed effect within their connected set. Two validations support the classification. Shrinkage falls by close to 7 percent over the nine months after a high-type manager arrives at a store, and rises by about as much after she leaves (Figure 1). And the managers who reduce shrinkage also carry better fixed effects on margins they do not record; their effects on turnover and on non-justified absence, which payroll and the time clock record, comove with their effect on shrinkage, with correlations between 0.38 and 0.82 (Figure 2). These managers are not better bookkeepers; they appear to manage differently. When a high-type manager arrives, she is in the store more: she works about 1.5 more shifts a month, including more weekend and night shifts, and her co-presence with the frontline rises by about 3 percentage points (Figure 3). She reworks the team she inherits, moving workers up a tier and across areas, while turnover rises through quits rather than dismissals (Figure 4). She does not add security or supervisory staff, and on demographic characteristics high-type managers are indistinguishable from the rest.

I then use the firm's policy of setting wages centrally by position, which makes a worker's premium move with the outside wage of her municipality, to relate shrinkage to the wage premium. Shrinkage is lower where the premium is higher: a 10 percent lower premium is associated with about 1.2 percent more shrinkage. Because the two levers do not cost the same, whether they act as complements or substitutes matters for how a firm should use them; if they are substitutes, a firm can lean on the cheaper one, while if they are complements, the manager needs a rent to threaten and the rent needs a monitor. The evidence points to complements. After a high-type arrival, shrinkage falls by close to 12 percent where the premium is high against roughly 4 percent where it is low; a further reduction of about eight percent. Where the premium is high, high-type managers also tend to be more present, although those differences are imprecisely estimated, and they enforce more: suspensions rise by 0.31 per hundred workers after a high-type arrival in a high-premium store and fall by 0.14 in a low-premium one, which is a significant difference at five percent.

Finally, I use Mexico's January minimum-wage hikes. Both statutory floors rise by the same percentage, so the step in Mexican pesos is larger at the border, and the firm's pay scale does not move with the floor, so the premium over the legal floor narrows more in border stores. Comparing border with interior stores of the same format around each hike, border shrinkage is unchanged in the first months and about 4.5 percent higher in the months that follow. The rise appears in stores whose manager is seldom on the floor and not in those where she is often present, a pattern the design shows but does not estimate precisely.

I read these facts through a shirking model of care (Shapiro and Stiglitz, 1984). The firm pays a fixed wage by position, so the only formal incentive is dismissal: a worker caught neglecting her job loses the premium she earns over her outside option. Whether she is caught depends on the manager's presence on the floor. The deterrent is therefore a product, the probability that a lapse is detected times the rent lost upon dismissal, and the product delivers the three facts. A better monitor disciplines care; a larger premium disciplines it too; and, under conditions the model states, the two are complements, so a given manager accomplishes more where the premium is high. Because the sanction is the manager's own act, the model also predicts that she sanctions where the job is worth keeping, which is where the suspensions rise.

For firms, the two levers are one decision: how much to pay above the market and whom to put in charge of a store are best chosen together. On the shrinkage margin alone, the premium is an expensive lever. In the average store, with about 67 workers and a premium of about USD 93 a month each, a 10 percent raise costs about USD 625 a month and is associated with about USD 180 less merchandise lost, about thirty cents per dollar. A high-type manager, in contrast, costs the firm what a low-type one costs, because pay is set by position; the cost of better management is the cost of identifying and developing managers, which at least in theory is lower than the cost of raising the wage of every worker in the store. And the same manager is worth more where the premium is high. The further reduction of 8.1 percent between high- and low-premium stores amounts to about USD 1,400 of merchandise a month in the average store; obtaining the same additional reduction through pay alone would take, by the premium gradient, a premium about 110 percent higher, more than double, equivalent to about USD 6,900 a month. Identifying, developing, and placing good managers is therefore a way to make an existing pay scale go further, and a pay scale set without regard to who manages each store leaves part of its return unrealized.

For policy, the minimum-wage findings suggest that a hike changes more than the wage bill. By narrowing the premium over the legal floor, it weakens one of the incentives a firm uses to sustain care, and the cost appears months later, on a margin that standard datasets do not record. How large that cost is depends on how the firm is organized; in this firm

the rise in shrinkage appeared where the manager was seldom on the floor. Firms facing a hike may therefore need to adjust how they manage, not only how much they pay, and evaluations that look only at employment and wages would miss that adjustment.

The paper contributes first to the evidence on how pay and monitoring interact. Coviello et al. (2022) establish a wage-by-monitoring complementarity: a minimum-wage increase raises the productivity of bound workers, and by more where monitoring is intensive. This paper finds the same complementarity from the opposite direction, moving the monitor rather than the wage: the same manager accomplishes more where the rent is large. It also measures the monitor differently, as an individual manager whose arrivals and presence on the floor are observed, rather than as a staffing ratio. Second, it adds an outcome: shrinkage, treated in economics as an accounting line, is read here as a measure of care and linked—store by store and manager by manager—to the personnel of the store. Third, it adds variation to the evidence that pay premia sustain discipline within firms (Cappelli and Chauvin, 1991; Emanuel and Harrington, 2026): the premium moves both with local labor markets and by statute, against the same audited outcome. Finally, it adds to the literature that recovers manager effects from rotation and asks what good managers do (Bertrand and Schoar, 2003; Lazear et al., 2015; Minni, 2026; Adhvaryu et al., 2026), by estimating the manager's effect on a margin a supervisor observes only imperfectly and showing that it depends on the wage environment in which the manager operates. The same reading is available wherever a firm audits a loss it cannot trace to anyone: orders remade, food thrown out, parts scrapped.

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